

Supplementary Information

Blast freezing a black hole

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S1. Microscopic model and evaporation protocol

The main text and Methods describe the quench schematically as initially decoupled η and χ SYK sectors that merge at t_{ev} . Here we give the microscopic realization of that shorthand model, including the relative normalization of the pure and mixed interactions and a slightly more general formation-and-evaporation protocol. Using the parameter s defined in Methods, the one-sided Hamiltonian is

$$\begin{aligned} H[\eta, \chi; J] = & i^{p/2} s^{-p/2} l_\eta(t) \sum_{I_p} J_{I_p} \eta_{I_p} + i^{p/2} (1-s)^{-p/2} l_\chi(t) \sum_{K_p} J_{K_p} \chi_{K_p} \\ & + i^{p/2} l_c(t) \sum_{m=1}^{p-1} \sum_{I_m, K_{p-m}} J_{I_m; K_{p-m}} \eta_{I_m} \chi_{K_{p-m}}. \end{aligned} \quad (\text{S1})$$

Here $p/2$ is even, matching the reflected-copy convention used below. The multi-index $I_m = (i_1, \dots, i_m)$ is an increasing m -tuple of η -indices, and $\eta_{I_m} \equiv \eta_{i_1} \cdots \eta_{i_m}$. Similarly, $K_n = (k_1, \dots, k_n)$ is an increasing n -tuple of χ -indices, with $\chi_{K_n} \equiv \chi_{k_1} \cdots \chi_{k_n}$.

All couplings associated with distinct index sets are independent Gaussian random variables with zero mean and common variance

$$\begin{aligned} \overline{J_{I_p}^2} = \overline{J_{K_p}^2} = \overline{J_{I_m; K_{p-m}}^2} &= J^2 \frac{(p-1)!}{(N_\eta + N_\chi)^{p-1}} \\ &= \mathcal{J}^2 \frac{2^{p-1}}{p} \frac{(p-1)!}{(N_\eta + N_\chi)^{p-1}}, \quad 1 \leq m \leq p-1. \end{aligned} \quad (\text{S2})$$

The left and right copies use the same disorder realization.

A sharp evaporation protocol that realizes the shorthand Hamiltonian in Methods is

$$\begin{aligned} l_\eta(t) &= a s^{1/2} \theta(t_{\text{ev}} - t) + s^{p/2} \theta(t - t_{\text{ev}}), \\ l_\chi(t) &= b (1-s)^{1/2} \theta(t_{\text{ev}} - t) + (1-s)^{p/2} \theta(t - t_{\text{ev}}), \\ l_c(t) &= \theta(t - t_{\text{ev}}). \end{aligned} \quad (\text{S3})$$

For $t < t_{\text{ev}}$, $l_c = 0$, while the total coefficients of the pure η and χ interactions are $a s^{(1-p)/2}$ and $b (1-s)^{(1-p)/2}$, respectively. Together with Eq. (S2), these give the standard SYK normalization for systems of sizes N_η and N_χ , with interaction scales controlled by a and b ; the main calculation uses $a = b = 1$. For $t > t_{\text{ev}}$, $s^{-p/2} l_\eta = (1-s)^{-p/2} l_\chi = l_c = 1$, so the pure and mixed p -body monomials have the same variance and assemble into $H_{\text{SYK}}[\eta \oplus \chi]$.

The left-right bilinear couplings in the doubled Hamiltonian can be parameterized more generally as

$$\begin{aligned} \mu_\eta(t) &= \mu [c \theta(-t) + \nu \theta(t - t_{\text{ev}})], \\ \mu_\chi(t) &= \mu [d \theta(t_{\text{ev}} - t) + \theta(t - t_{\text{ev}})]. \end{aligned} \quad (\text{S4})$$

Here c, d, ν are dimensionless constants. The evaporation protocol used in the main calculation is $c = \nu = 0$, $d = 1$, and $\mu = \mu_0$, for which $\mu_\eta(t) = 0$ and $\mu_\chi(t) = \mu_0$. The formation-and-evaporation extension discussed in the main text and Methods instead takes $c \neq 0$, $\nu = 0$, and $d = 1$: the η bilinear is present for $t < 0$, is switched off at $t = 0$, and the η - χ interaction is subsequently switched on at t_{ev} . The more general choices $d \neq 1$ and $\nu \neq 0$ would allow, respectively, a change of the bath bilinear at t_{ev} and an η bilinear after evaporation.

More generally, the changes in $l_\eta, l_\chi, l_c, \mu_\eta$, and μ_χ may occur at distinct times, and the step functions may be replaced by smooth profiles, including the gradual evaporation mentioned in the main text and Methods. The calculations presented here use the sharp profiles above.

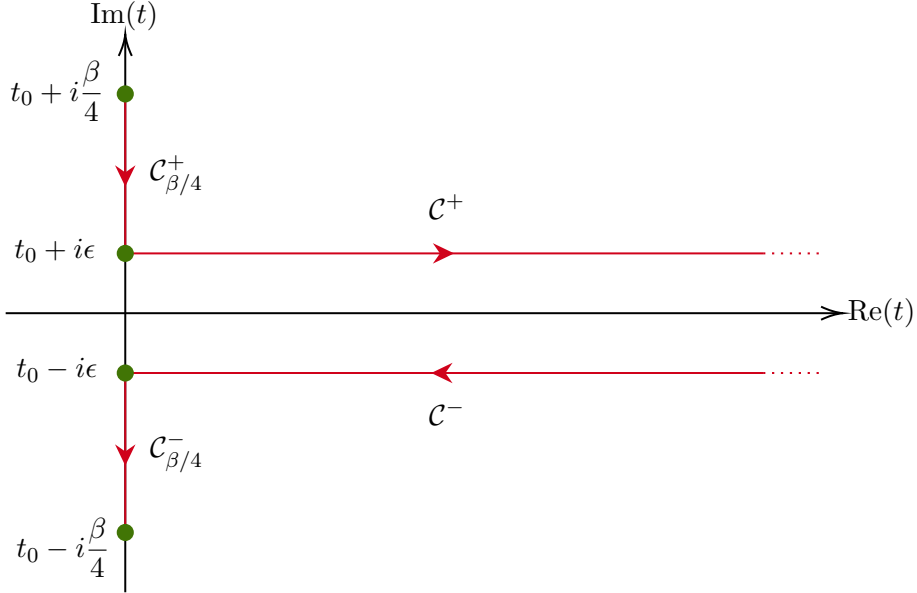


Figure S1: **Contour for real-time evolution and TFD preparation.** The horizontal branches $\mathcal{C}^+$ and $\mathcal{C}^-$ describe forward and backward real-time evolution, while $\mathcal{C}_{\beta/4}^+$ and $\mathcal{C}_{\beta/4}^-$ prepare the initial thermofield-double data. The endpoint conditions are given in Eq. (S9).

S2. Bilocal effective theory

Methods summarizes the large- N and large- p equations used in the main text. Here we derive those equations from the disorder-averaged bilocal action and spell out the contour conventions.

Large- N scaling and real-time contour. We first take $N_\eta, N_\chi \rightarrow \infty$ at fixed s and fixed p . With the normalization introduced in Sec. S1, the decoupled η and χ sectors contribute at order N_η and N_χ , respectively, whereas the post-evaporation action is of order $N_\eta + N_\chi$. Throughout this section, $a, b \in \{R, L\}$ denote copy indices and should not be confused with the protocol constants a and b in Sec. S1.

The calculation is performed on the Schwinger–Keldysh contour shown in Fig. S1, defined by

$$\mathcal{C} = \mathcal{C}_{\beta/4}^+ \cup \mathcal{C}^+ \cup \mathcal{C}^- \cup \mathcal{C}_{\beta/4}^-. \quad (\text{S5})$$

The forward and backward real-time branches $\mathcal{C}^+$ and $\mathcal{C}^-$ compute in-in observables, while the vertical segments prepare the common-inverse-temperature β thermofield-double data used in the finite- p calculation of Sec. S5. The analytic solutions in Sec. S3 instead allow independent initial data for the two decoupled sectors: a thermal η state and a stationary coupled χ bath. The real-time equations below apply to both preparations, with the corresponding Euclidean boundary data.

For each species $\psi = \eta, \chi$, define

$$c_j^\psi = \frac{\psi_j^L + i\psi_j^R}{\sqrt{2}}, \quad (c_j^\psi)^\dagger = \frac{\psi_j^L - i\psi_j^R}{\sqrt{2}}. \quad (\text{S6})$$

The reference state $|\mathbf{1}\rangle$ is annihilated by all c_j^ψ . Equivalently,

$$\psi_j^L |\mathbf{1}\rangle = -i\psi_j^R |\mathbf{1}\rangle, \quad \langle \mathbf{1} | \psi_j^L = i \langle \mathbf{1} | \psi_j^R. \quad (\text{S7})$$

The thermofield-double state is prepared by Euclidean evolution,

$$|\text{TFD}\rangle = \frac{1}{\sqrt{Z_\beta}} e^{-\frac{\beta}{4}(H^R + H^L)} |\mathbf{1}\rangle. \quad (\text{S8})$$

This representation fixes the endpoint conditions

$$\begin{aligned}\psi_j^L(t_0 + i\beta/4) &= -i\psi_j^R(t_0 + i\beta/4), \\ \psi_j^L(t_0 - i\beta/4) &= i\psi_j^R(t_0 - i\beta/4).\end{aligned}\tag{S9}$$

Define the contour-ordered two-point function by

$$G_{ab}^\psi(z_1, z_2) = -i \langle \mathcal{T}_C \psi_a(z_1) \psi_b(z_2) \rangle. \tag{S10}$$

The field-level conditions in Eq. (S9) imply, for example,

$$\begin{aligned}G_{aL}^\psi(z, t_0 + i\beta/4) &= -iG_{aR}^\psi(z, t_0 + i\beta/4), \\ G_{La}^\psi(t_0 - i\beta/4, z) &= iG_{Ra}^\psi(t_0 - i\beta/4, z).\end{aligned}\tag{S11}$$

These relations provide the thermal boundary data for the real-time evolution.

Disorder-averaged effective action. At the path-integral level, introduce the microscopic collective fields

$$\hat{G}_{ab}^\psi(z_1, z_2) = -\frac{i}{N_\psi} \sum_{j=1}^{N_\psi} \psi_j^a(z_1) \psi_j^b(z_2), \quad \psi = \eta, \chi. \tag{S12}$$

Independent bilocal variables $\mathcal{G}^\psi$ are constrained to equal $\hat{G}^\psi$, with Σ^ψ acting as the corresponding Lagrange multipliers. This is done by inserting the identity

$$\begin{aligned}1 = \int D\Sigma^\eta D\Sigma^\chi D\mathcal{G}^\eta D\mathcal{G}^\chi \exp \left[-\frac{(N_\eta + N_\chi)}{2} \int_C dz_1 dz_2 \sum_{a,b} \left\{ s \Sigma_{ab}^\eta (\mathcal{G}_{ab}^\eta - \hat{G}_{ab}^\eta) \right. \right. \\ \left. \left. + (1-s) \Sigma_{ab}^\chi (\mathcal{G}_{ab}^\chi - \hat{G}_{ab}^\chi) \right\} \right].\end{aligned}\tag{S13}$$

The factors s and $1-s$ appear because $N_\eta = s(N_\eta + N_\chi)$ and $N_\chi = (1-s)(N_\eta + N_\chi)$. After imposing the constraint, we drop the calligraphic notation; the saddle value G^ψ is the contour correlator defined in Eq. (S10).

After disorder averaging and imposing the bilocal constraint, the fermion path integral is Gaussian. For a Majorana quadratic form,

$$\int D\psi \exp \left[\frac{i}{2} \int_C dz_1 dz_2 \psi(z_1) (G_0^{-1} - \Sigma) (z_1, z_2) \psi(z_2) \right] = \exp \left[\frac{1}{2} \text{Tr} \log [-i (G_0^{-1} - \Sigma)] \right]. \tag{S14}$$

Applying this to the η and χ fermions gives the two trace-log terms in the effective action.

The interaction terms follow from the Gaussian disorder average

$$\int \frac{dJ}{\sqrt{2\pi \langle J^2 \rangle}} \exp \left[-\frac{J^2}{2 \langle J^2 \rangle} + JA \right] = \exp \left[\frac{\langle J^2 \rangle}{2} A^2 \right]. \tag{S15}$$

At leading order in N_ψ , the sum over increasing p -tuples gives

$$\sum_{I_p} \psi_{I_p}^a(z_1) \psi_{I_p}^b(z_2) \simeq \frac{N_\psi^p}{p!} \left(G_{ab}^\psi(z_1, z_2) \right)^p. \tag{S16}$$

The disorder average of the mixed interactions produces the binomial combination

$$\begin{aligned}l_c(z_1) l_c(z_2) \sum_{m=1}^{p-1} \binom{p}{m} s^m (1-s)^{p-m} (2G_{ab}^\eta)^m (2G_{ab}^\chi)^{p-m} \\ = l_c(z_1) l_c(z_2) \left[(2[sG_{ab}^\eta + (1-s)G_{ab}^\chi])^p - s^p (2G_{ab}^\eta)^p - (1-s)^p (2G_{ab}^\chi)^p \right].\end{aligned}\tag{S17}$$

The last two terms subtract the pure $m = p$ and $m = 0$ interactions, which are already included separately.

After integrating out the fermions, the disorder-averaged partition function takes the saddle form

$$\overline{Z} = \int DG D\Sigma \exp [i(N_\eta + N_\chi)S[G, \Sigma]]. \quad (\text{S18})$$

The effective action is

$$\begin{aligned} S[G, \Sigma] = & -s \frac{i}{2} \text{Tr} \log [-i (G_0^{-1} - \Sigma^\eta)] - (1-s) \frac{i}{2} \text{Tr} \log [-i (G_0^{-1} - \Sigma^\chi)] \\ & + \frac{i}{2} \int_{\mathcal{C}} dz_1 dz_2 \sum_{a,b} [s \Sigma_{ab}^\eta G_{ab}^\eta + (1-s) \Sigma_{ab}^\chi G_{ab}^\chi] \\ & - \frac{i \mathcal{J}^2}{4p^2} \int_{\mathcal{C}} dz_1 dz_2 \sum_{a,b} (-1)^{1+p\delta_{ab}/2} \left[l_\eta(z_1) l_\eta(z_2) (2G_{ab}^\eta)^p \right. \\ & \quad + l_\chi(z_1) l_\chi(z_2) (2G_{ab}^\chi)^p \\ & \quad \left. + l_c(z_1) l_c(z_2) \left\{ (2[sG_{ab}^\eta + (1-s)G_{ab}^\chi])^p \right. \right. \\ & \quad \left. \left. - s^p (2G_{ab}^\eta)^p - (1-s)^p (2G_{ab}^\chi)^p \right\} \right] \\ & + \frac{1}{2p} \int_{\mathcal{C}} dz_1 dz_2 \delta_{\mathcal{C}}(z_1, z_2) \left[s \mu_\eta(z_1) (G_{LR}^\eta - G_{RL}^\eta) \right. \\ & \quad \left. + (1-s) \mu_\chi(z_1) (G_{LR}^\chi - G_{RL}^\chi) \right]. \end{aligned} \quad (\text{S19})$$

Here Tr includes the trace over the R, L indices and integration along the contour. All bilocals in Eq. (S19) are evaluated at (z_1, z_2) unless their arguments are displayed explicitly. The factor $(-1)^{1+p\delta_{ab}/2}$ incorporates the Majorana phases and the common-disorder reflected-copy convention used in Methods. The contour coupling $\mu_\psi(z)$ restricts to the protocol in Eq. (S4) on the real-time branches and vanishes on the Euclidean preparation segments.

The bare inverse propagator is

$$(G_0^{-1})_{ab}(z_1, z_2) = i \delta_{ab} \partial_{z_1} \delta_{\mathcal{C}}(z_1, z_2). \quad (\text{S20})$$

Varying Eq. (S19) with respect to G^ψ gives

$$\begin{aligned} \Sigma_{ab}^\psi(z_1, z_2) = & (-1)^{1+p\delta_{ab}/2} \frac{\mathcal{J}^2}{p} \left[\alpha_\psi(z_1, z_2) (2G_{ab}^\psi(z_1, z_2))^{p-1} \right. \\ & \left. + l_c(z_1) l_c(z_2) (2[sG_{ab}^\eta(z_1, z_2) + (1-s)G_{ab}^\chi(z_1, z_2)])^{p-1} \right] \\ & - i \epsilon_{ab} \frac{\mu_\psi(z_1)}{p} \delta_{\mathcal{C}}(z_1, z_2), \end{aligned} \quad (\text{S21})$$

where

$$\epsilon := \begin{pmatrix} \epsilon_{RR} & \epsilon_{RL} \\ \epsilon_{LR} & \epsilon_{LL} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad (\text{S22})$$

and

$$\begin{aligned} \alpha_\eta(z_1, z_2) &= s^{-1} l_\eta(z_1) l_\eta(z_2) - s^{p-1} l_c(z_1) l_c(z_2), \\ \alpha_\chi(z_1, z_2) &= (1-s)^{-1} l_\chi(z_1) l_\chi(z_2) - (1-s)^{p-1} l_c(z_1) l_c(z_2). \end{aligned} \quad (\text{S23})$$

For the sharp protocol in Eq. (S3), when both contour arguments lie before evaporation one has $\alpha_\eta = a^2$, $\alpha_\chi = b^2$, and $l_c = 0$. When both arguments lie after evaporation, $\alpha_\eta = \alpha_\chi = 0$ and $l_c = 1$, leaving only the common self-energy built from $sG^\eta + (1-s)G^\chi$. This reproduces the independent pre-evaporation saddles and the combined post-evaporation saddle. Expanding the latter term gives the weighted sum of products $G_\eta^k G_\chi^{p-1-k}$ summarized schematically in the main text and Methods.

Kadanoff-Baym equations. In this paragraph G^ψ and Σ^ψ denote 2×2 matrices in the R, L copy indices. Matrix multiplication in this space and convolution along $\mathcal{C}$ are implicit.

Varying with respect to Σ^ψ gives the contour Dyson equation

$$\Sigma^\psi(z_1, z_2) = G_0^{-1}(z_1, z_2) - \left(G^\psi\right)^{-1}(z_1, z_2), \quad (\text{S24})$$

where the inverse is defined by

$$\int_{\mathcal{C}} dz_3 \left(G^\psi\right)^{-1}(z_1, z_3) G^\psi(z_3, z_2) = \delta_{\mathcal{C}}(z_1, z_2) \mathbb{I}. \quad (\text{S25})$$

Multiplying Eq. (S24) by G^ψ from the right or left gives the two contour Kadanoff-Baym equations,

$$\begin{aligned} i\partial_{z_1} G^\psi(z_1, z_2) &= \delta_{\mathcal{C}}(z_1, z_2) \mathbb{I} + \int_{\mathcal{C}} dz_3 \Sigma^\psi(z_1, z_3) G^\psi(z_3, z_2), \\ -i\partial_{z_2} G^\psi(z_1, z_2) &= \delta_{\mathcal{C}}(z_1, z_2) \mathbb{I} + \int_{\mathcal{C}} dz_3 G^\psi(z_1, z_3) \Sigma^\psi(z_3, z_2). \end{aligned} \quad (\text{S26})$$

The two equations are related by complex conjugation. With our conventions, complex conjugation reverses the orientation of the contour. In components, $(i)^* = -i$, $\delta_{\mathcal{C}}(z_1, z_2)^* = -\delta_{\mathcal{C}}(z_1^*, z_2^*)$, and $(\int_{\mathcal{C}} dz)^* = -\int_{\mathcal{C}} dz^*$. The bilocals obey

$$\left(G_{ab}^\psi(z_1, z_2)\right)^* = -G_{ba}^\psi(z_2^*, z_1^*), \quad \left(\Sigma_{ab}^\psi(z_1, z_2)\right)^* = -\Sigma_{ba}^\psi(z_2^*, z_1^*).$$

Taking the complex conjugate of the first contour Eq. (S26) and then relabeling $z_1 \leftrightarrow z_2$ gives the second equation.

For real times, define

$$\begin{aligned} G_{ab}^{\psi, >}(t_1, t_2) &= -i \langle \psi_a(t_1) \psi_b(t_2) \rangle, \\ G_{ab}^{\psi, <}(t_1, t_2) &= i \langle \psi_b(t_2) \psi_a(t_1) \rangle = \left(G_{ab}^{\psi, >}(t_1, t_2)\right)^*. \end{aligned} \quad (\text{S27})$$

The smooth interaction self-energy obeys the analogous relation $\Sigma^{\psi, <} = (\Sigma^{\psi, >})^*$, component by component. Superscripts R, A below denote retarded and advanced components and should not be confused with the R, L copy labels.

Applying the Langreth rules to the first contour KB equation gives

$$\begin{aligned} i\partial_{t_1} G^{\psi, >}(t_1, t_2) &= \int_{t_0}^{\infty} dt_3 \left[\Sigma^{\psi, \text{R}}(t_1, t_3) G^{\psi, >}(t_3, t_2) + \Sigma^{\psi, >}(t_1, t_3) G^{\psi, \text{A}}(t_3, t_2) \right] \\ &\quad + \int_{\mathcal{C}_{\beta/4}^+ \cup \mathcal{C}_{\beta/4}^-} dz_3 \Sigma^\psi(t_1, z_3) G^\psi(z_3, t_2). \end{aligned} \quad (\text{S28})$$

The required retarded and advanced components are

$$\begin{aligned} G^{\psi, \text{R}}(t_1, t_2) &= \theta(t_1 - t_2) \left[G^{\psi, >}(t_1, t_2) - G^{\psi, <}(t_1, t_2) \right], \\ G^{\psi, \text{A}}(t_1, t_2) &= -\theta(t_2 - t_1) \left[G^{\psi, >}(t_1, t_2) - G^{\psi, <}(t_1, t_2) \right], \\ \Sigma^{\psi, \text{R}}(t_1, t_2) &= \theta(t_1 - t_2) \left[\Sigma^{\psi, >}(t_1, t_2) - \Sigma^{\psi, <}(t_1, t_2) \right] - i \frac{\mu_\psi(t_1)}{p} \delta(t_1 - t_2) \epsilon, \\ \Sigma^{\psi, \text{A}}(t_1, t_2) &= -\theta(t_2 - t_1) \left[\Sigma^{\psi, >}(t_1, t_2) - \Sigma^{\psi, <}(t_1, t_2) \right] - i \frac{\mu_\psi(t_1)}{p} \delta(t_1 - t_2) \epsilon. \end{aligned} \quad (\text{S29})$$

For real t_1, t_2 , the Wightman components obey

$$G^{\psi, <}(t_1, t_2) = \left(G^{\psi, >}(t_1, t_2)\right)^*, \quad \Sigma^{\psi, <}(t_1, t_2) = \left(\Sigma^{\psi, >}(t_1, t_2)\right)^*, \quad (\text{S30})$$

where complex conjugation is taken componentwise. Therefore

$$\begin{aligned}\Sigma^{\psi,R}(t_1, t_2) &= 2i \theta(t_1 - t_2) \operatorname{Im} \Sigma^{\psi,>}(t_1, t_2) - i \frac{\mu_\psi(t_1)}{p} \delta(t_1 - t_2) \epsilon, \\ G^{\psi,A}(t_1, t_2) &= -2i \theta(t_2 - t_1) \operatorname{Im} G^{\psi,>}(t_1, t_2).\end{aligned}\tag{S31}$$

Substituting these relations into Eq. (S28) yields

$$\begin{aligned}\partial_{t_1} G^{\psi,>}(t_1, t_2) &= 2 \int_{t_0}^{t_1} dt_3 \operatorname{Im} [\Sigma^{\psi,>}(t_1, t_3)] G^{\psi,>}(t_3, t_2) \\ &\quad - 2 \int_{t_0}^{t_2} dt_3 \Sigma^{\psi,>}(t_1, t_3) \operatorname{Im} [G^{\psi,>}(t_3, t_2)] \\ &\quad - \frac{\mu_\psi(t_1)}{p} \epsilon G^{\psi,>}(t_1, t_2) - i \int_{\mathcal{C}_{\beta/4}^+ \cup \mathcal{C}_{\beta/4}^-} dz_3 \Sigma^\psi(t_1, z_3) G^\psi(z_3, t_2).\end{aligned}\tag{S32}$$

Here $\Sigma^{\psi,>}$ in the real-time memory integrals denotes the smooth interaction part of the self-energy; the local bilinear contribution has already been written separately. The last line,

$$\mathcal{I}_{\text{th}}^\psi(t_1, t_2) = -i \int_{\mathcal{C}_{\beta/4}^+ \cup \mathcal{C}_{\beta/4}^-} dz_3 \Sigma^\psi(t_1, z_3) G^\psi(z_3, t_2),\tag{S33}$$

contains the thermal-preparation data and generally depends on both t_1 and t_2 . The companion equation with a derivative acting on t_2 follows from the second contour KB equation.

Large- p simplification. We now take the large- N , large- p limit with

$$\frac{p^2}{N_\eta + N_\chi} \longrightarrow 0,\tag{S34}$$

while keeping s finite at this stage. For the reflection-symmetric matrix form used here, we further take $p/2$ even. The reflection, conjugation, and normalization conventions are collected in Sec. S6. The probe limit and its analytic two-point functions are treated separately in Sec. S3.

Using the large- p parametrization of $G^{\psi,>}$ introduced in the main text, define

$$\mathcal{K}_a^\psi(t_1, t_2) = \alpha_\psi(t_1, t_2) e^{g_a^\psi(t_1, t_2)} + l_c(t_1) l_c(t_2) e^{s g_a^\eta(t_1, t_2) + (1-s) g_a^\chi(t_1, t_2)}, \quad a = R, L.\tag{S35}$$

The smooth interaction part of the greater self-energy becomes

$$\Sigma^{\psi,>}(t_1, t_2) = -\frac{\mathcal{J}^2}{p} \begin{pmatrix} i\mathcal{K}_R^\psi(t_1, t_2) & -\mathcal{K}_L^\psi(t_1, t_2) \\ \mathcal{K}_L^\psi(t_1, t_2) & i\mathcal{K}_R^\psi(t_1, t_2) \end{pmatrix}.\tag{S36}$$

The local bilinear term is excluded from Eq. (S36) and appears explicitly in Eq. (S32).

At leading order in $1/p$, the first real-time KB equation gives

$$\begin{aligned}\partial_{t_1} g_R^\psi(t_1, t_2) &= -2\mathcal{J}^2 \int_{t_0}^{t_1} dt_3 \left[\operatorname{Re} \mathcal{K}_R^\psi(t_1, t_3) - i \operatorname{Im} \mathcal{K}_L^\psi(t_1, t_3) \right] \\ &\quad + 2\mathcal{J}^2 \int_{t_0}^{t_2} dt_3 \mathcal{K}_R^\psi(t_1, t_3) - i\mu_\psi(t_1) + \mathcal{V}_R^\psi(t_1), \\ \partial_{t_1} g_L^\psi(t_1, t_2) &= -2\mathcal{J}^2 \int_{t_0}^{t_1} dt_3 \left[\operatorname{Re} \mathcal{K}_R^\psi(t_1, t_3) - i \operatorname{Im} \mathcal{K}_L^\psi(t_1, t_3) \right] \\ &\quad - 2\mathcal{J}^2 \int_{t_0}^{t_2} dt_3 \mathcal{K}_L^\psi(t_1, t_3) - i\mu_\psi(t_1) + \mathcal{V}_L^\psi(t_1).\end{aligned}\tag{S37}$$

The one-time functions generated by the Euclidean preparation segments are

$$\begin{aligned}
\mathcal{V}_R^\psi(t_1) &= \mathcal{J}^2 \int_{\mathcal{C}_{\beta/4}^+} dz_3 \left[\mathcal{K}_R^\psi(t_1, z_3) + \mathcal{K}_L^\psi(t_1, z_3) \right] \\
&\quad + \mathcal{J}^2 \int_{\mathcal{C}_{\beta/4}^-} dz_3 \left[\mathcal{K}_R^\psi(z_3, t_1) + \mathcal{K}_L^\psi(z_3, t_1) \right], \\
\mathcal{V}_L^\psi(t_1) &= -\mathcal{J}^2 \int_{\mathcal{C}_{\beta/4}^+} dz_3 \left[\mathcal{K}_R^\psi(t_1, z_3) + \mathcal{K}_L^\psi(t_1, z_3) \right] \\
&\quad + \mathcal{J}^2 \int_{\mathcal{C}_{\beta/4}^-} dz_3 \left[\mathcal{K}_R^\psi(z_3, t_1) + \mathcal{K}_L^\psi(z_3, t_1) \right].
\end{aligned} \tag{S38}$$

Here $\mathcal{K}_{R,L}^\psi$ is analytically continued to the Euclidean segments where necessary.

Unlike the finite- p thermal term $\mathcal{I}_{\text{th}}^\psi(t_1, t_2)$, its leading large- p projections $\mathcal{V}_{R,L}^\psi(t_1)$ depend only on t_1 . Taking a t_2 derivative of Eq. (S37) therefore gives the local Liouville-type equations

$$\begin{aligned}
\partial_{t_1} \partial_{t_2} g_R^\psi(t_1, t_2) &= 2\mathcal{J}^2 \mathcal{K}_R^\psi(t_1, t_2), \\
\partial_{t_1} \partial_{t_2} g_L^\psi(t_1, t_2) &= -2\mathcal{J}^2 \mathcal{K}_L^\psi(t_1, t_2).
\end{aligned} \tag{S39}$$

The companion first-order equations with a derivative acting on t_2 follow by complex conjugation and exchange of the two time arguments.

The thermal gluing conditions become

$$g_L^\psi(t, t_0 + i\beta/4) = g_R^\psi(t, t_0 + i\beta/4), \quad g_L^\psi(t_0 - i\beta/4, t) = g_R^\psi(t_0 - i\beta/4, t). \tag{S40}$$

The first-order equations also supply boundary data that are lost upon taking the second derivative. The ultraviolet normalization $g_R^\psi(t, t) = 0$ implies

$$(\partial_{t_1} + \partial_{t_2}) g_R^\psi(t_1, t_2) \Big|_{t_1=t_2=t} = 0, \tag{S41}$$

while the off-diagonal component obeys

$$(\partial_{t_1} - \partial_{t_2}) g_L^\psi(t_1, t_2) \Big|_{t_1=t_2=t} = -2i\mu_\psi(t). \tag{S42}$$

At a discontinuous quench, these equal-time relations are understood as one-sided limits. Thus the local Liouville equations must be supplemented by the thermal gluing conditions and the first-order KB boundary data.

S3. Probe limit and analytic two-point functions

Probe limit and reduced kernels. We now specialize the large- p equations of Sec. S2 to the probe limit used in the main text and Methods,

$$p \longrightarrow \infty, \quad s \longrightarrow 0, \quad sp \longrightarrow 0, \quad \frac{p^2}{N_\eta} \longrightarrow 0, \quad b = d = 1. \tag{S43}$$

The small-sector condition $p^2/N_\eta \rightarrow 0$ controls the pre-evaporation η SYK expansion even as $s \rightarrow 0$, and also implies $p^2/(N_\eta + N_\chi) \rightarrow 0$. The χ sector is therefore unaffected by the η sector, whereas the η self-energy after evaporation is determined by the fixed χ background. We retain the parameters a and ν defined in Sec. S1; the main calculation uses $a = b = d = 1$ and $c = \nu = 0$. The formulas below assume thermal η data at $t = 0$. A formation stage with $c \neq 0$ can supply these data at leading large p when its initial wormhole is tuned to the thermal angle specified below; a general formation history requires its own matching data.

For $t \geq 0$, the protocol in Sec. S1 reduces to

$$\begin{aligned} l_\eta(t) &\longrightarrow a s^{1/2} \theta(t_{\text{ev}} - t), & l_\chi(t) &\longrightarrow 1, \\ l_c(t) &= \theta(t - t_{\text{ev}}), & \mu_\eta(t) &= \nu \mu_0 \theta(t - t_{\text{ev}}), \\ \mu_\chi(t) &= \mu_0. \end{aligned} \quad (\text{S44})$$

It is convenient to denote the post-evaporation difference of the bilinear couplings by

$$\Delta\mu \equiv \mu_\chi - \mu_\eta = \mu_0(1 - \nu). \quad (\text{S45})$$

Equation (S23) gives

$$\begin{aligned} \alpha_\eta(t_1, t_2) &= a^2 \theta(t_{\text{ev}} - t_1) \theta(t_{\text{ev}} - t_2), \\ \alpha_\chi(t_1, t_2) &= 1 - \theta(t_1 - t_{\text{ev}}) \theta(t_2 - t_{\text{ev}}). \end{aligned} \quad (\text{S46})$$

Consequently, the kernels in Eq. (S35) become

$$\begin{aligned} \mathcal{K}_X^\chi(t_1, t_2) &= e^{g_X^\chi(t_1, t_2)}, \\ \mathcal{K}_X^\eta(t_1, t_2) &= a^2 e^{g_X^\eta(t_1, t_2)} \theta(t_{\text{ev}} - t_1) \theta(t_{\text{ev}} - t_2) \\ &\quad + e^{g_X^\chi(t_1, t_2)} \theta(t_1 - t_{\text{ev}}) \theta(t_2 - t_{\text{ev}}), \quad X = R, L. \end{aligned} \quad (\text{S47})$$

In particular, $\mathcal{K}_X^\eta(t_1, t_2) = 0$ when the two arguments lie on opposite sides of t_{ev} . For the main-text choice $a = 1$, substitution into Eq. (S39) gives Eqs. (5) and (6).

Bath and pre-evaporation solutions. The fixed χ background is the time-translation-invariant traversable-wormhole solution [1, 2]. Define $0 < r_G < 1$, ϕ_G , and V_G by

$$\begin{aligned} r_G^2 &= e^{-2\phi_G} = \frac{\mu_0}{2\mathcal{J}} \left[\sqrt{1 + \left(\frac{\mu_0}{4\mathcal{J}} \right)^2} - \frac{\mu_0}{4\mathcal{J}} \right], \\ V_G &= \frac{r_G}{\sqrt{1 - r_G^2}}. \end{aligned} \quad (\text{S48})$$

Equivalently,

$$\mu_0 = 2\mathcal{J} \frac{e^{-2\phi_G}}{\sqrt{1 - e^{-2\phi_G}}}.$$

The bath correlators are

$$\begin{aligned} e^{g_R^\chi(t_1, t_2)} &= e^{-i\pi} \left[\frac{V_G}{\sin(\mathcal{J}V_G(t_1 - t_2) - i \tanh^{-1} r_G)} \right]^2, \\ e^{g_L^\chi(t_1, t_2)} &= \left[\frac{V_G}{\cos(\mathcal{J}V_G(t_1 - t_2) - i \tanh^{-1} r_G)} \right]^2. \end{aligned} \quad (\text{S49})$$

These expressions obey $g_R^\chi(t, t) = 0$ and $g_L^\chi(t, t) = -2\phi_G$. Whenever $g_{R,L}^\chi$ rather than its exponential appears below, the continuous logarithm is chosen consistently with these equal-time values and the conjugation relations of Sec. S6. The Lensky-Qi parametrization underlying this solution is also reviewed there. This stationary bath is chosen independently of the η temperature. At leading large p , its initial correlators agree with TFD data tuned to $\vartheta_\chi = \arcsin r_G$ and $\beta_\chi \mathcal{J} r_G = \pi - 2 \arcsin r_G$; in general $\beta_\chi \neq \beta_\eta$. The β in the pre-evaporation η solution below denotes β_η . The common-temperature finite- p preparation of Sec. S5 is evolved numerically and is not assumed to be an exact stationary bath saddle.

Before evaporation, with the η bilinear turned off, the solution for $0 < t_1, t_2 < t_{\text{ev}}$ is

$$\begin{aligned} e^{g_R^\eta(t_1, t_2)} &= e^{-i\pi} \left[\frac{\sin \vartheta}{\sinh(a\mathcal{J} \sin \vartheta (t_1 - t_2) - i\vartheta)} \right]^2, \\ e^{g_L^\eta(t_1, t_2)} &= \left[\frac{\sin \vartheta}{\cosh(a\mathcal{J} \sin \vartheta (t_1 + t_2))} \right]^2. \end{aligned} \quad (\text{S50})$$

The thermal angle $0 < \vartheta < \pi/2$ is fixed by

$$\beta a \mathcal{J} \sin \vartheta = \pi - 2\vartheta. \quad (\text{S51})$$

We use ϑ , rather than ϵ , to distinguish this thermal parameter from the antisymmetric R, L matrix in Eq. (S22).

Matching across the evaporation surface. Consider first $t_2 < t_{\text{ev}} < t_1$. Because the mixed-region η kernel vanishes, the local Liouville equation is homogeneous there. The first-order KB equation (S37) fixes the two single-time functions that are left undetermined by the second-order equation. The result is

$$g_X^\eta(t_1, t_2) = g_X^\eta(t_{\text{ev}}, t_2) + \text{Re } g_R^\chi(t_1, t_{\text{ev}}) + i \text{Im } g_L^\chi(t_1, t_{\text{ev}}) + i \Delta\mu(t_1 - t_{\text{ev}}), \quad X = R, L. \quad (\text{S52})$$

The single-time functions f_{1X} and f_{2X} appearing in Eq. (7) can therefore be read off directly from Eq. (S52).

The region $t_1 < t_{\text{ev}} < t_2$ follows from

$$g_X^\eta(t_1, t_2) = [g_X^\eta(t_2, t_1)]^*. \quad (\text{S53})$$

Equation (S52) makes the factorization of the mixed-region correlator explicit:

$$\left| e^{g_X^\eta(t_1, t_2)} \right| = \left| e^{g_X^\eta(t_{\text{ev}}, t_2)} \right| \left| e^{g_R^\chi(t_1, t_{\text{ev}})} \right|, \quad X = R, L. \quad (\text{S54})$$

The same factorization holds with every exponent replaced by g/p , as in the physical Wightman functions. Since the bath correlator is bounded in the traversable-wormhole background, i.e. $\left| e^{g_R^\chi(t_1, t_{\text{ev}})} \right| \leq 1$, we obtain

$$\left| e^{g_X^\eta(t_1, t_2)} \right| \leq \left| e^{g_X^\eta(t_{\text{ev}}, t_2)} \right|, \quad X = R, L. \quad (\text{S55})$$

Thus the simple η correlator cannot grow in magnitude across the evaporation surface.

Post-evaporation solution. For $t_1, t_2 > t_{\text{ev}}$, the probe kernel is sourced entirely by the bath,

$$\mathcal{K}_X^\eta(t_1, t_2) = e^{g_X^\chi(t_1, t_2)}, \quad X = R, L. \quad (\text{S56})$$

The second-order Liouville equation determines $\partial_{t_1} \partial_{t_2} g_X^\eta$, but by itself leaves undetermined functions of the individual time arguments. These functions are fixed by the first-order KB equation. In the leading large- p probe equations, the KB evolution is causal in its first time argument and can be integrated from $t_1 = t_{\text{ev}}$ in the form

$$g_X^\eta(t_1, t_2) = g_X^\eta(t_{\text{ev}}, t_2) + \mathcal{F}_X^\chi(t_1, t_2), \quad \mathcal{F}_X^\chi(t_{\text{ev}}, t_2) = 0, \quad X = R, L, \quad (\text{S57})$$

where $\mathcal{F}_X^\chi$ is completely determined by the known bath correlator and the prescribed post-evaporation bilinear couplings.

Setting $t_2 = t_{\text{ev}}$ in Eq. (S57), and then using the real-time conjugation relation in Eq. (S53), gives

$$\begin{aligned} g_X^\eta(t_1, t_{\text{ev}}) &= g_X^\eta(t_{\text{ev}}, t_{\text{ev}}) + \mathcal{F}_X^\chi(t_1, t_{\text{ev}}), \\ g_X^\eta(t_{\text{ev}}, t_2) &= g_X^\eta(t_{\text{ev}}, t_{\text{ev}}) + [\mathcal{F}_X^\chi(t_2, t_{\text{ev}})]^*. \end{aligned} \quad (\text{S58})$$

Here we used the reality of the equal-time values. Substitution back into Eq. (S57) yields

$$g_X^\eta(t_1, t_2) = g_X^\eta(t_{\text{ev}}, t_{\text{ev}}) + \mathcal{F}_X^\chi(t_1, t_2) + [\mathcal{F}_X^\chi(t_2, t_{\text{ev}})]^*. \quad (\text{S59})$$

Thus no independent function on the matching surface is required: the first-order evolution and conjugation reduce the η matching data to their coincident values at $(t_{\text{ev}}, t_{\text{ev}})$. This closure is a feature

of the leading large- p , probe-limit equations used here and would not follow from the second-order Liouville equation alone.

Evaluating the bath-dependent functional explicitly, define

$$\begin{aligned} f_{3R}(t) &= i \operatorname{Im} [-g_R^\chi(t, t_{\text{ev}}) + g_L^\chi(t, t_{\text{ev}})] + i\Delta\mu(t - t_{\text{ev}}), \\ f_{3L}(t) &= \frac{1}{2} [g_L^\eta(t_{\text{ev}}, t_{\text{ev}}) + g_L^\chi(t_{\text{ev}}, t_{\text{ev}})] + \operatorname{Re} [g_R^\chi(t, t_{\text{ev}}) - g_L^\chi(t, t_{\text{ev}})] + i\Delta\mu(t - t_{\text{ev}}). \end{aligned} \quad (\text{S60})$$

The complete post-evaporation solution is therefore

$$g_X^\eta(t_1, t_2) = g_X^\chi(t_1, t_2) + f_{3X}(t_1) + [f_{3X}(t_2)]^*, \quad X = R, L, \quad (\text{S61})$$

which is the explicit form of Eq. (8). The required equal-time matching data follow from the pre-evaporation solution:

$$g_R^\eta(t_{\text{ev}}, t_{\text{ev}}) = 0, \quad \left| e^{g_L^\eta(t_{\text{ev}}, t_{\text{ev}})} \right| = \left[\frac{\sin \vartheta}{\cosh(2a\mathcal{J}t_{\text{ev}} \sin \vartheta)} \right]^2. \quad (\text{S62})$$

Since f_{3R} is purely imaginary,

$$\left| e^{g_R^\eta(t_1, t_2)} \right| = \left| e^{g_R^\chi(t_1, t_2)} \right|. \quad (\text{S63})$$

Post-evaporation left-right bound. For the left-right component, Eq. (S61) gives

$$\left| e^{g_L^\eta(t_1, t_2)} \right| = h_\chi(t_1, t_2; t_{\text{ev}}) \left| e^{g_L^\eta(t_{\text{ev}}, t_{\text{ev}})} \right|, \quad (\text{S64})$$

where the nonnegative bath-dependent transfer factor is

$$h_\chi(t_1, t_2; t_{\text{ev}}) \equiv \left| e^{g_L^\chi(t_1, t_2) + g_L^\chi(t_{\text{ev}}, t_{\text{ev}})} \times \frac{e^{g_R^\chi(t_1, t_{\text{ev}}) + g_R^\chi(t_{\text{ev}}, t_2)}}{e^{g_L^\chi(t_1, t_{\text{ev}}) + g_L^\chi(t_{\text{ev}}, t_2)}} \right|. \quad (\text{S65})$$

Post-evaporation bound: Suppose that the bath correlators admit the normalized physical Lensky-Qi parametrization with a common differentiable function $y(t)$ satisfying

$$\operatorname{Im} y(t) < 0, \quad |y'(t)| = \sinh(2\mathcal{J}|\operatorname{Im} y(t)|), \quad (\text{S66})$$

and that the post-evaporation matching form (S61) applies. Then

$$0 \leq h_\chi(t_1, t_2; t_{\text{ev}}) \leq 1, \quad t_1, t_2 \geq t_{\text{ev}}. \quad (\text{S67})$$

The proof is given in Sec. S6 using the disk representation $w(t) = e^{-2i\mathcal{J}y(t)}$. The disk inequality does not require the bath trajectory to be a fixed point or to be time-translation invariant; its scope is the normalized single-function Lensky-Qi class, conditional on the post-evaporation matching form derived above. The phase proportional to the real parameter $\Delta\mu$ drops out of the modulus.

For the stationary bath used in the main text, define $x_i = \mathcal{J}V_G(t_i - t_{\text{ev}})$. Direct substitution of Eq. (S49) gives

$$h_\chi(t_1, t_2; t_{\text{ev}}) = \frac{(V_G^2 + \cos^2 x_1)(V_G^2 + \cos^2 x_2)}{(V_G^2 + \sin^2 x_1)(V_G^2 + \sin^2 x_2)} \times \frac{V_G^4}{[V_G^2 + \cos^2(x_1 - x_2)](V_G^2 + 1)} \leq 1. \quad (\text{S68})$$

The non-strict inequality is essential: it is saturated at $t_1 = t_2 = t_{\text{ev}}$, and whenever both disk points $w(t_1)$ and $w(t_2)$ return to $w(t_{\text{ev}})$. A single argument on the matching surface does not in general saturate the bound.

Combining Eqs. (S64) and (S67), we obtain

$$\left| e^{g_L^\eta(t_1, t_2)} \right| \leq \left| e^{g_L^\eta(t_{\text{ev}}, t_{\text{ev}})} \right|, \quad t_1, t_2 \geq t_{\text{ev}}. \quad (\text{S69})$$

Since $p > 0$, the corresponding physical Wightman factors obey

$$\left| e^{g_L^\eta(t_1, t_2)/p} \right| \leq \left| e^{g_L^\eta(t_{\text{ev}}, t_{\text{ev}})/p} \right|. \quad (\text{S70})$$

Thus the post-evaporation left-right correlator cannot exceed its value on the evaporation surface. Its dependence on the pre-evaporation history is carried entirely by this matching value, which becomes parametrically small at late t_{ev} according to Eq. (S62).

Anticommutator matrix. For bulk reconstruction it is useful to package the simple-channel anti-commutators into a 2×2 matrix. Flavor symmetry gives

$$A_{aj,bk}(t_1, t_2) = \left\langle \left\{ \eta_j^a(t_1), \eta_k^b(t_2) \right\} \right\rangle = \delta_{jk} A_{ab}(t_1, t_2), \quad a, b \in \{R, L\}. \quad (\text{S71})$$

In terms of Wightman functions,

$$\begin{aligned} A_{ab}(t_1, t_2) &= i \left[G_{ab}^{\eta, >}(t_1, t_2) + G_{ba}^{\eta, >}(t_2, t_1) \right] \\ &= i \left[G_{ab}^{\eta, >}(t_1, t_2) - (G_{ab}^{\eta, >}(t_1, t_2))^* \right] = -2 \operatorname{Im} G_{ab}^{\eta, >}(t_1, t_2). \end{aligned} \quad (\text{S72})$$

Using the large- p ansatz,

$$A_{RR}(t_1, t_2) = \operatorname{Re} \left[e^{g_R^\eta(t_1, t_2)/p} \right], \quad A_{RL}(t_1, t_2) = \operatorname{Im} \left[e^{g_L^\eta(t_1, t_2)/p} \right]. \quad (\text{S73})$$

Together with $A_{LL} = A_{RR}$ and $A_{LR} = -A_{RL}$, this gives

$$A(t_1, t_2) = \begin{pmatrix} \operatorname{Re} \left[e^{g_R^\eta(t_1, t_2)/p} \right] & \operatorname{Im} \left[e^{g_L^\eta(t_1, t_2)/p} \right] \\ -\operatorname{Im} \left[e^{g_L^\eta(t_1, t_2)/p} \right] & \operatorname{Re} \left[e^{g_R^\eta(t_1, t_2)/p} \right] \end{pmatrix}. \quad (\text{S74})$$

The conjugation properties of g_R^η and g_L^η imply

$$A_{RR}(t_2, t_1) = A_{RR}(t_1, t_2), \quad A_{RL}(t_2, t_1) = -A_{RL}(t_1, t_2). \quad (\text{S75})$$

This anticommutator matrix supplies the boundary inner product used in the bulk reconstruction summarized in Eq. (26).

S4. Operator size calculation

Methods states the folded-contour response used to compute ΔL_η and ΔL_χ . Here we derive the response functions and list their region-by-region forms.

We study the fate of a simple η excitation created before evaporation and ask how it is detected by the size operators in the η and χ sectors after evaporation. We use $\psi \in \{\eta, \chi\}$ to denote the sector whose size operator is measured.

Using the creation and annihilation operators defined in Eq. (S6), the single-mode number operator is

$$\hat{n}_j^\psi = (c_j^\psi)^\dagger c_j^\psi = \frac{1}{2} + i\psi_j^L \psi_j^R. \quad (\text{S76})$$

Thus

$$\hat{\mathcal{N}}_\psi = \sum_{j=1}^{N_\psi} \hat{n}_j^\psi = \frac{N_\psi}{2} + \hat{L}_\psi, \quad \hat{L}_\psi = i \sum_{j=1}^{N_\psi} \psi_j^L \psi_j^R. \quad (\text{S77})$$

The constant $N_\psi/2$ cancels in differences of expectation values, so $\Delta \mathcal{N}_\psi = \Delta L_\psi$.

Let $|\Phi\rangle$ denote the unperturbed state. A normalized simple η excitation inserted from the right boundary at $0 < t_b < t_{\text{ev}}$ is

$$|\Phi_{b,j}\rangle = \sqrt{2} \eta_j^R(t_b) |\Phi\rangle. \quad (\text{S78})$$

Its background-subtracted size at a measurement time $t_0 > t_{\text{ev}}$ is

$$\Delta L_\psi^{\text{bdry}}(t_b; t_0) = \frac{1}{N_\eta} \sum_{j=1}^{N_\eta} \left[\langle \Phi_{b,j} | \hat{L}_\psi(t_0) | \Phi_{b,j} \rangle - \langle \Phi | \hat{L}_\psi(t_0) | \Phi \rangle \right]. \quad (\text{S79})$$

The normalization is by N_η , because the inserted fermion belongs to the η sector.

To compute this response, introduce the RR generating correlator

$$f_\psi(t_1, t_2; t_0; \lambda) = -\frac{2i}{N_\eta} \sum_{j=1}^{N_\eta} \langle \Phi | e^{i\lambda \hat{L}_\psi(t_0)/p} \eta_j^R(t_1) e^{-i\lambda \hat{L}_\psi(t_0)/p} \eta_j^R(t_2) | \Phi \rangle. \quad (\text{S80})$$

For equal insertion times,

$$\Delta L_{\psi}^{\text{bdry}}(t_b; t_0) = -p \partial_{\lambda} f_{\psi}(t_b, t_b; t_0; \lambda)|_{\lambda=0}. \quad (\text{S81})$$

The conjugations in Eq. (S80) are represented by folding the real-time contour at t_0 . On the unfolded contour the Hamiltonian is

$$H_{\lambda}^{(\psi)}(t) = \theta(t_0 - t)H(t) - \theta(t - t_0)H(2t_0 - t) + \frac{\lambda}{p}\delta(t - t_0)\hat{L}_{\psi}. \quad (\text{S82})$$

For a physical time $t_1 < t_0$ on the return branch, define

$$\tilde{t}_1 = 2t_0 - t_1, \quad \tilde{t}_{\text{ev}} = 2t_0 - t_{\text{ev}}. \quad (\text{S83})$$

When the Lensky–Qi formula is expressed in unfolded contour times, its right-hand side includes the fixed orientation factor $\varsigma(u)\varsigma(v)$, where $\varsigma = +1$ on the outward branch and $\varsigma = -1$ on the return branch. This compensates the reversed derivative of the folded trajectory and preserves the equal-physical-time normalization. It is independent of λ and therefore cancels from all logarithmic responses below. Thus t_1, t_2 below denote physical boundary times, whereas $\tilde{t}_1 > t_0 > t_2$ is the corresponding independent contour ordering. In this ordering,

$$f_{\psi}(t_1, t_2; t_0; \lambda) = 2G_{RR,\psi}^{\eta,>}(\tilde{t}_1, t_2; t_0, \lambda), \quad (\text{S84})$$

where the subscript ψ records the sector in which the source is inserted; the external fermions remain η fermions. Using

$$G_{RR,\psi}^{\eta,>}(\tilde{t}_1, t_2; t_0, \lambda) = -\frac{i}{2} \exp \left[\frac{g_{R,\psi}^{\eta}(\tilde{t}_1, t_2; t_0, \lambda)}{p} \right], \quad (\text{S85})$$

define

$$\begin{aligned} g_{R,0}^{\eta}(t_1, t_2) &\equiv g_{R,\psi}^{\eta}(\tilde{t}_1, t_2; t_0, 0), \\ \mathfrak{D}_{\psi}(t_1, t_2; t_0) &\equiv i \partial_{\lambda} g_{R,\psi}^{\eta}(\tilde{t}_1, t_2; t_0, \lambda) \Big|_{\lambda=0}. \end{aligned} \quad (\text{S86})$$

It follows that the RR bilocal response is

$$\Delta L_{\psi}(t_1, t_2; t_0) = \exp \left[\frac{g_{R,0}^{\eta}(t_1, t_2)}{p} \right] \mathfrak{D}_{\psi}(t_1, t_2; t_0). \quad (\text{S87})$$

For unequal t_1, t_2 , this is a folded-contour bilocal response rather than an individual real size expectation value. For $t_1 = t_2 = t_b$, the unperturbed folded evolution cancels and $g_{R,0}^{\eta}(t_b, t_b) = 0$. Therefore

$$\Delta L_{\psi}^{\text{bdry}}(t_b; t_0) = \mathfrak{D}_{\psi}(t_b, t_b; t_0). \quad (\text{S88})$$

Boundary data at the reflected evaporation surface. We first determine the data that the kick at t_0 induces at $\tilde{t}_{\text{ev}}$. The natural variables are the Lensky–Qi variables reviewed in Sec. S6. In particular, Eqs. (S220) and (S222) give

$$y'(t) = \frac{e^{ip(t)}}{\sqrt{e^{2\phi(t)} - 1}}, \quad \mathcal{H}_Q = -2\mathcal{J}_{\text{eff}} \sqrt{1 - e^{-2\phi}} \cos p + \mu_{\text{eff}}(t)\phi. \quad (\text{S89})$$

Here, as in Sec. S6, $p(t)$ is the Lensky–Qi canonical phase and should not be confused with the interaction order p .

The delta-function source in Eq. (S82) shifts the bilinear coupling and hence produces the canonical jump

$$p(t_0^+) - p(t_0^-) = -\lambda. \quad (\text{S90})$$

Writing $\delta \equiv \partial_{\lambda}|_{\lambda=0}$, the directly kicked sector therefore obeys

$$\delta\phi(t_0) = 0, \quad \delta p(t_0) = -1. \quad (\text{S91})$$

The post- t_0 segment transports this kick to $\tilde{t}_{\text{ev}}$. We denote the matching data induced in the pre-evaporation η trajectory by $\delta\phi_\psi^{\text{eff}}(\tilde{t}_{\text{ev}})$ and $\delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}})$. The superscript “eff” distinguishes these data from the actual bath fluctuation produced by a χ -sector kick. Both sources give

$$\delta\phi_\psi^{\text{eff}}(\tilde{t}_{\text{ev}}) = 0, \quad \delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}}) \in \mathbb{R}, \quad (\text{S92})$$

with

$$\begin{aligned} \delta p_\eta^{\text{eff}}(\tilde{t}_{\text{ev}}) &= -1, \\ \delta p_\chi^{\text{eff}}(\tilde{t}_{\text{ev}}) &= \frac{1 - r_G^2}{2 - r_G^2} [1 - \cos(\omega_G(t_0 - t_{\text{ev}}))]. \end{aligned} \quad (\text{S93})$$

Here

$$\omega_G^2 = 4\mathcal{J}^2 \frac{r_G^2(2 - r_G^2)}{1 - r_G^2}, \quad (\text{S94})$$

and r_G, ϕ_G , and V_G are defined in Eq. (S48). We derive Eqs. (S92) and (S93) below.

Pre-evaporation propagation. The effect of the post-evaporation source is summarized by the effective data at $\tilde{t}_{\text{ev}}$, while propagation from t_{ev} back to the insertion time is a purely pre-evaporation η problem. This step is the same for the η - and χ -size sources; the only difference is the value of $\delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}})$.

Set

$$\mathcal{J}_\eta \equiv a\mathcal{J}. \quad (\text{S95})$$

In terms of the thermal angle ϑ defined in Eq. (S51), the unperturbed pre-evaporation trajectory is

$$y_{0,\eta}(t) = \frac{1}{\mathcal{J}_\eta} \tan^{-1} \tanh\left(\mathcal{J}_\eta t \sin \vartheta - \frac{i\vartheta}{2}\right), \quad (\text{S96})$$

in agreement with Eq. (S224). The linearized solution generated by a ψ -sector source can be written as

$$\delta_\psi y_\eta(t) = \delta\gamma_\psi + y'_{0,\eta}(t) \left[-\delta t_{\star,\psi} + \delta\vartheta_\psi \frac{2\mathcal{J}_\eta t \cos \vartheta - i}{2\mathcal{J}_\eta \sin \vartheta} \right]. \quad (\text{S97})$$

These three terms are obtained by varying the additive y -shift, time origin, and thermal angle of the exact trajectory, and hence solve the linearized Liouville equation. Imposing the data in Eq. (S92), together with coordinate continuity $\delta_\psi y_\eta(t_{\text{ev}}) = 0$, fixes all three constants and gives

$$\begin{aligned} \delta\vartheta_\psi &= \delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}}) \tanh(2\mathcal{J}_\eta t_{\text{ev}} \sin \vartheta), \\ \delta t_{\star,\psi} &= \delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}}) \cot \vartheta \left[t_{\text{ev}} \tanh(2\mathcal{J}_\eta t_{\text{ev}} \sin \vartheta) - \frac{\csc \vartheta}{2\mathcal{J}_\eta} \right], \\ \delta\gamma_\psi &= -\frac{\delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}})}{2\mathcal{J}_\eta} \csc \vartheta \operatorname{sech}(2\mathcal{J}_\eta t_{\text{ev}} \sin \vartheta). \end{aligned} \quad (\text{S98})$$

For two physical times $t_1, t_2 < t_{\text{ev}}$, variation of the Lensky–Qi form in Eq. (S216) gives

$$\mathfrak{D}_\psi^{\text{pre}}(t_1, t_2; t_0) = i \left[\frac{\delta_\psi y'_\eta(t_1)}{y'_{0,\eta}(t_1)} - 2\mathcal{J}_\eta \delta_\psi y_\eta(t_1) \cot(\mathcal{J}_\eta [y_{0,\eta}(t_1) - y_{0,\eta}(t_2)^*]) \right]. \quad (\text{S99})$$

All times on the right-hand side are reflected physical times; this convention accounts for the orientation of the return branch.

For the equal-time boundary insertion, substituting Eq. (S98) into Eq. (S99) gives

$$\mathfrak{D}_\psi(t_b, t_b; t_0) = -\delta p_\psi^{\text{eff}}(\tilde{t}_{\text{ev}}) \mathcal{E}(t_b, t_{\text{ev}}), \quad (\text{S100})$$

where

$$\begin{aligned} \mathcal{E}(t_b, t_{\text{ev}}) &= 1 - \csc^2 \vartheta \left[1 - \frac{\cosh(2\mathcal{J}_\eta t_b \sin \vartheta)}{\cosh(2\mathcal{J}_\eta t_{\text{ev}} \sin \vartheta)} \right] \\ &\quad + 2\mathcal{J}_\eta (t_{\text{ev}} - t_b) \frac{\cos^2 \vartheta}{\sin \vartheta} \tanh(2\mathcal{J}_\eta t_{\text{ev}} \sin \vartheta). \end{aligned} \quad (\text{S101})$$

Propagating the kick to the reflected evaporation surface. We now derive the effective data quoted in Eq. (S93). For a source in the η sector, the bath is unperturbed in the probe limit. On the return branch, for $t_0 < u < \tilde{t}_{\text{ev}}$ and $t_2 < t_{\text{ev}}$, the mixed-region solution following from Eq. (S52) is

$$g_R^\eta(u, t_2; t_0, \lambda) = g_R^\eta(t_{\text{ev}}, t_2) + \text{Re } g_R^\chi(u, t_{\text{ev}}) + i \text{Im } g_L^\chi(u, t_{\text{ev}}) + i\Delta\mu(2t_0 - u - t_{\text{ev}}) - i\lambda, \quad (\text{S102})$$

where $\Delta\mu$ is defined in Eq. (S45). Setting $u = \tilde{t}_{\text{ev}}$ and $t_2 = t_{\text{ev}}$ gives

$$\partial_\lambda g_R^\eta(\tilde{t}_{\text{ev}}, t_{\text{ev}}; t_0, \lambda)|_{\lambda=0} = -i. \quad (\text{S103})$$

From Eq. (S89),

$$\frac{\delta y'}{y'} = i \delta p - \frac{\delta \phi}{1 - e^{-2\phi}}. \quad (\text{S104})$$

At the matching surface $\delta_\eta y_\eta(t_{\text{ev}}) = 0$ and $\delta \phi_\eta^{\text{eff}}(\tilde{t}_{\text{ev}}) = 0$. Hence Eq. (S103) implies

$$\delta p_\eta^{\text{eff}}(\tilde{t}_{\text{ev}}) = -1. \quad (\text{S105})$$

For a source in the χ sector, the kick first excites the bath. The unperturbed bath is the fixed point

$$\phi_\chi = \phi_G, \quad p_\chi = 0, \quad (\text{S106})$$

as in Eq. (S225). To distinguish the actual bath fluctuation from the effective pre-evaporation matching datum, write it as $(\delta \phi_\chi^{\text{bath}}, \delta p_\chi^{\text{bath}})$. Its initial data are

$$\delta \phi_\chi^{\text{bath}}(t_0) = 0, \quad \delta p_\chi^{\text{bath}}(t_0) = -1. \quad (\text{S107})$$

Linearizing the Hamilton equations in Eq. (S223) on the return branch gives

$$\begin{aligned} -\delta \dot{\phi}_\chi^{\text{bath}} &= 2\mathcal{J}\sqrt{1 - r_G^2} \delta p_\chi^{\text{bath}}, \\ -\delta \dot{p}_\chi^{\text{bath}} &= -2\mathcal{J}\frac{r_G^2(2 - r_G^2)}{(1 - r_G^2)^{3/2}} \delta \phi_\chi^{\text{bath}}. \end{aligned} \quad (\text{S108})$$

Equivalently, $\delta \ddot{\phi}_\chi^{\text{bath}} + \omega_G^2 \delta \phi_\chi^{\text{bath}} = 0$, with ω_G as in Eq. (S94). The solution is

$$\begin{aligned} \delta \phi_\chi^{\text{bath}}(t) &= \frac{2\mathcal{J}\sqrt{1 - r_G^2}}{\omega_G} \sin[\omega_G(t - t_0)], \\ \delta p_\chi^{\text{bath}}(t) &= -\cos[\omega_G(t - t_0)], \quad t > t_0. \end{aligned} \quad (\text{S109})$$

Substituting this bath fluctuation into the linearized mixed-region matching at $t = \tilde{t}_{\text{ev}}$ gives

$$\begin{aligned} \delta \phi_\chi^{\text{eff}}(\tilde{t}_{\text{ev}}) &= 0, \\ \delta p_\chi^{\text{eff}}(\tilde{t}_{\text{ev}}) &= \frac{1 - r_G^2}{2 - r_G^2} [1 - \cos(\omega_G(t_0 - t_{\text{ev}}))], \end{aligned} \quad (\text{S110})$$

which completes the derivation of Eq. (S93).

Combining Eqs. (S88), (S100), (S105), and (S110), one obtains

$$\begin{aligned} \Delta L_\eta^{\text{bdry}}(t_b; t_0) &= \mathcal{E}(t_b, t_{\text{ev}}), \\ \Delta L_\chi^{\text{bdry}}(t_b; t_0) &= -\frac{1 - r_G^2}{2 - r_G^2} [1 - \cos(\omega_G(t_0 - t_{\text{ev}}))] \mathcal{E}(t_b, t_{\text{ev}}). \end{aligned} \quad (\text{S111})$$

The negative χ response is consistent with this definition: it is a change relative to the background size, rather than an absolute occupation number.

Bilocal size response in all regions. We now keep the two physical insertion times $t_1, t_2 < t_0$ independent. The generating correlator remains Eq. (S80), and on the unfolded contour we work in the independent ordering

$$\tilde{t}_1 = 2t_0 - t_1 > t_0 > t_2. \quad (\text{S112})$$

The corresponding response is given by Eq. (S87). The exponential prefactor equals one for $t_1 = t_2 = t_b$, but is nontrivial for a general bilocal.

Linearized bath solution. For a χ -sector source we require the variation of the bath functions that enter the RR probe solution. The unperturbed folded bath trajectory is

$$y_{0,\chi}^{\text{fold}}(t) = \begin{cases} V_G t - \frac{i}{2\mathcal{J}} \tanh^{-1}(r_G), & t < t_0, \\ V_G(2t_0 - t) - \frac{i}{2\mathcal{J}} \tanh^{-1}(r_G), & t > t_0. \end{cases} \quad (\text{S113})$$

For $t > t_0$, the kick-induced variation of y is

$$\delta_\chi y_{\text{bath}}(t) = \frac{iV_G}{\omega_G} \sin[\omega_G(t - t_0)] + \frac{2\mathcal{J}V_G}{\omega_G^2 \sqrt{1 - r_G^2}} [1 - \cos(\omega_G(t - t_0))]. \quad (\text{S114})$$

For $u > t_0 > v$, define

$$\mathcal{B}_X(u, v; t_0) \equiv \partial_\lambda g_X^\chi(u, v; t_0, \lambda)|_{\lambda=0}, \quad X = R, L. \quad (\text{S115})$$

The Lensky–Qi parametrization gives

$$\begin{aligned} \mathcal{B}_R(u, v; t_0) &= \frac{\delta_\chi y'_{\text{bath}}(u)}{\left(y_{0,\chi}^{\text{fold}}\right)'(u)} - 2\mathcal{J} \delta_\chi y_{\text{bath}}(u) \cot\left(\mathcal{J} \left[y_{0,\chi}^{\text{fold}}(u) - y_{0,\chi}^{\text{fold}}(v)^*\right]\right), \\ \mathcal{B}_L(u, v; t_0) &= \frac{\delta_\chi y'_{\text{bath}}(u)}{\left(y_{0,\chi}^{\text{fold}}\right)'(u)} + 2\mathcal{J} \delta_\chi y_{\text{bath}}(u) \tan\left(\mathcal{J} \left[y_{0,\chi}^{\text{fold}}(u) - y_{0,\chi}^{\text{fold}}(v)^*\right]\right). \end{aligned} \quad (\text{S116})$$

Here $\left(y_{0,\chi}^{\text{fold}}\right)'(u) = -V_G$ for $u > t_0$. The quantity $\mathcal{B}_L$ is not an RL size response: it is an internal bath function required because the RR probe solution in Sec. S3 contains $\text{Im } g_L^\chi$. For an η -sector source the bath is unperturbed in the probe limit, and both bath variations vanish.

$t_1, t_2 < t_{\text{ev}}$. Both physical insertion times lie before evaporation. The response is

$$\mathfrak{D}_\psi(t_1, t_2; t_0) = \mathfrak{D}_\psi^{\text{pre}}(t_1, t_2; t_0), \quad t_1, t_2 < t_{\text{ev}}, \quad (\text{S117})$$

where the right-hand side is given in Eq. (S99), with the parameters in Eq. (S98).

$t_2 < t_{\text{ev}} < t_1 < t_0$. The mixed-region solution is

$$\begin{aligned} g_R^\eta(\tilde{t}_1, t_2; t_0, \lambda) &= g_R^\eta(t_{\text{ev}}, t_2) + \text{Re } g_R^\chi(\tilde{t}_1, t_{\text{ev}}; t_0, \lambda) + i \text{Im } g_L^\chi(\tilde{t}_1, t_{\text{ev}}; t_0, \lambda) \\ &\quad + i\Delta\mu(t_1 - t_{\text{ev}}) + \begin{cases} -i\lambda, & \psi = \eta, \\ +i\lambda, & \psi = \chi. \end{cases} \end{aligned} \quad (\text{S118})$$

Therefore

$$\begin{aligned} \mathfrak{D}_\eta(t_1, t_2; t_0) &= 1, \\ \mathfrak{D}_\chi(t_1, t_2; t_0) &= -1 + i \text{Re } \mathcal{B}_R(\tilde{t}_1, t_{\text{ev}}; t_0) - \text{Im } \mathcal{B}_L(\tilde{t}_1, t_{\text{ev}}; t_0), \end{aligned} \quad t_2 < t_{\text{ev}} < t_1 < t_0. \quad (\text{S119})$$

$t_{\text{ev}} < t_1, t_2 < t_0$. When both physical insertion times lie after evaporation,

$$\begin{aligned}
g_R^\eta(\tilde{t}_1, t_2; t_0, \lambda) &= g_R^\chi(\tilde{t}_1, t_2; t_0, \lambda) \\
&+ i \operatorname{Im} \left[-g_R^\chi(\tilde{t}_1, t_{\text{ev}}; t_0, \lambda) - g_R^\chi(t_{\text{ev}}, t_2) + g_L^\chi(\tilde{t}_1, t_{\text{ev}}; t_0, \lambda) + g_L^\chi(t_{\text{ev}}, t_2) \right] \\
&+ i \Delta\mu(t_1 - t_2) + \begin{cases} -i\lambda, & \psi = \eta, \\ +i\lambda, & \psi = \chi. \end{cases}
\end{aligned} \tag{S120}$$

The terms whose two arguments lie before t_0 are independent of λ . Hence

$$\begin{aligned}
\mathfrak{D}_\eta(t_1, t_2; t_0) &= 1, \\
\mathfrak{D}_\chi(t_1, t_2; t_0) &= -1 + i \mathcal{B}_R(\tilde{t}_1, t_2; t_0) \quad t_{\text{ev}} < t_1, t_2 < t_0. \\
&+ \operatorname{Im} \left[\mathcal{B}_R(\tilde{t}_1, t_{\text{ev}}; t_0) - \mathcal{B}_L(\tilde{t}_1, t_{\text{ev}}; t_0) \right],
\end{aligned} \tag{S121}$$

$t_1 < t_{\text{ev}} < t_2 < t_0$. In this ordering, differentiation of the t_2 -evolution equation gives

$$\partial_{t_2} \left[\partial_\lambda g_R^\eta(\tilde{t}_1, t_2; t_0, \lambda) \Big|_{\lambda=0} \right] = \partial_{t_2} \left[\partial_\lambda g_R^\chi(\tilde{t}_{\text{ev}}, t_2; t_0, \lambda) \Big|_{\lambda=0} \right]. \tag{S122}$$

Integrating from t_{ev} to t_2 yields

$$\begin{aligned}
\mathfrak{D}_\eta(t_1, t_2; t_0) &= \mathfrak{D}_\eta^{\text{pre}}(t_1, t_{\text{ev}}; t_0), \\
\mathfrak{D}_\chi(t_1, t_2; t_0) &= \mathfrak{D}_\chi^{\text{pre}}(t_1, t_{\text{ev}}; t_0) \quad t_1 < t_{\text{ev}} < t_2 < t_0. \\
&+ i \left[\mathcal{B}_R(\tilde{t}_{\text{ev}}, t_2; t_0) - \mathcal{B}_R(\tilde{t}_{\text{ev}}, t_{\text{ev}}; t_0) \right],
\end{aligned} \tag{S123}$$

Summary. The direct η -sector response is

$$\mathfrak{D}_\eta(t_1, t_2; t_0) = \begin{cases} \mathfrak{D}_\eta^{\text{pre}}(t_1, t_2; t_0), & t_1, t_2 < t_{\text{ev}}, \\ 1, & t_2 < t_{\text{ev}} < t_1 < t_0, \\ 1, & t_{\text{ev}} < t_1, t_2 < t_0, \\ \mathfrak{D}_\eta^{\text{pre}}(t_1, t_{\text{ev}}; t_0), & t_1 < t_{\text{ev}} < t_2 < t_0. \end{cases} \tag{S124}$$

The χ -sector response is

$$\mathfrak{D}_\chi(t_1, t_2; t_0) = \begin{cases} \mathfrak{D}_\chi^{\text{pre}}(t_1, t_2; t_0), & t_1, t_2 < t_{\text{ev}}, \\ \text{the second line of Eq. (S119),} & t_2 < t_{\text{ev}} < t_1 < t_0, \\ \text{the second line of Eq. (S121),} & t_{\text{ev}} < t_1, t_2 < t_0, \\ \text{the second line of Eq. (S123),} & t_1 < t_{\text{ev}} < t_2 < t_0. \end{cases} \tag{S125}$$

Together with Eq. (S87), these formulas determine the independent folded-contour RR response in every physical time region. The physical boundary size is recovered by setting $t_1 = t_2 = t_b$. The response relevant to a reconstructed right-moving bulk mode is obtained by taking the corresponding linear superposition with the right-boundary reconstruction kernel in Eq. (26). This gives the operator-size response shown in Fig. 3.

S5. Two-replica calculation and Rényi mutual information

Methods summarizes the two-replica KB equation and the readout of $e^{-I^{(2)}}$ and $S^{(2)}$. Here we give the reduced-state identities, fermionic signs, numerical discretization, and on-shell-action formula.

Reduced-state overlaps and transition matrices. We now give the two-replica construction underlying the Rényi-2 mutual information in Eq. (14). The ordinary real-time evolution is performed on the folded contour $\mathcal{C}$ defined in Eq. (S5); the replica construction uses two copies of this contour. In this section $\alpha, \beta \in \{1, 2\}$ denote replica indices, while $a, b \in \{R, L\}$ denote the physical left-right copies, as in Sec. S2.

Let $|\Phi_{\text{in}}\rangle$ be the initial doubled state prepared by the Euclidean parts of the contour. The unperturbed state at the final replica-gluing time t_f is

$$|\Phi(t_f)\rangle = U(t_f, 0)|\Phi_{\text{in}}\rangle. \quad (\text{S126})$$

The boundary-fermion building block used below is the time-evolved version of the excitation in Eq. (S78),

$$|\Phi_{b,j}(t_f)\rangle = \sqrt{2} U(t_f, t_b) \eta_j^R U(t_b, 0) |\Phi_{\text{in}}\rangle, \quad 0 < t_b < t_{\text{ev}} < t_f. \quad (\text{S127})$$

The factor $\sqrt{2}$ gives unit norm because $(\eta_j^R)^2 = 1/2$. Define the corresponding reduced density matrices by

$$\begin{aligned} \rho_\eta(t_f) &= \text{Tr}_\chi [|\Phi(t_f)\rangle\langle\Phi(t_f)|], & \rho_\chi(t_f) &= \text{Tr}_\eta [|\Phi(t_f)\rangle\langle\Phi(t_f)|], \\ \sigma_{\eta,j}(t_b; t_f) &= \text{Tr}_\chi [|\Phi_{b,j}(t_f)\rangle\langle\Phi_{b,j}(t_f)|], & \sigma_{\chi,j}(t_b; t_f) &= \text{Tr}_\eta [|\Phi_{b,j}(t_f)\rangle\langle\Phi_{b,j}(t_f)|]. \end{aligned} \quad (\text{S128})$$

Because the unperturbed full state is pure, the two unperturbed reduced density matrices have the same nonzero Schmidt eigenvalues. Their purities are therefore equal:

$$\mathcal{P}_\eta(t_f) \equiv \text{Tr}_\eta \rho_\eta(t_f)^2 = \text{Tr}_\chi \rho_\chi(t_f)^2 \equiv \mathcal{P}_\chi(t_f). \quad (\text{S129})$$

The first normalized contraction required for the mutual-information calculation is

$$F_2(t_b; t_f) = \frac{1}{N_\eta} \sum_{j=1}^{N_\eta} \frac{\text{Tr}_\eta [\sigma_{\eta,j}(t_b; t_f) \rho_\eta(t_f)]}{\mathcal{P}_\eta(t_f)}. \quad (\text{S130})$$

It is convenient to express the numerator as a transition-matrix contraction in the bath Hilbert space. Define

$$\tau_{\chi,j}(t_b; t_f) = \text{Tr}_\eta [|\Phi_{b,j}(t_f)\rangle\langle\Phi(t_f)|]. \quad (\text{S131})$$

For an ordered fermionic occupation basis $\{|i_\eta, x_\chi\rangle\}$, write

$$(\Phi_{b,j})_{ix} = \langle i_\eta, x_\chi | \Phi_{b,j}(t_f) \rangle, \quad \Phi_{ix} = \langle i_\eta, x_\chi | \Phi(t_f) \rangle. \quad (\text{S132})$$

A direct contraction of these coefficients gives

$$\begin{aligned} \text{Tr}_\eta [\sigma_{\eta,j} \rho_\eta] &= \sum_{i,i',x,y} (\Phi_{b,j})_{ix} (\Phi_{b,j})_{i'x}^* \Phi_{i'y} \Phi_{iy}^* \\ &= \text{Tr}_\chi [\tau_{\chi,j}^\dagger \tau_{\chi,j}]. \end{aligned} \quad (\text{S133})$$

No additional fermion sign is generated in this regrouping: after an ordered occupation basis has been fixed, the wavefunction components are ordinary complex coefficients. Although $\tau_{\chi,j}$ is fermion odd, $\tau_{\chi,j}^\dagger \tau_{\chi,j}$ is fermion even. Consequently,

$$F_2(t_b; t_f) = \frac{\frac{1}{N_\eta} \sum_{j=1}^{N_\eta} \text{Tr}_\chi [\tau_{\chi,j}^\dagger(t_b; t_f) \tau_{\chi,j}(t_b; t_f)]}{\mathcal{P}_\chi(t_f)}. \quad (\text{S134})$$

In this representation the η indices close within each transition strip, while the χ indices are cyclically glued between the two strips. Thus the replica twist may equivalently be placed on the χ fields; its contour realization in the branch-dependent replica frame is shown in Fig. S2.

We choose replica 1 to represent $\tau_{\chi,j}^\dagger$ and replica 2 to represent $\tau_{\chi,j}$. The insertion from $\langle \Phi_{b,j}(t_f) |$ therefore lies on the lower branch of replica 1, whereas the insertion from $| \Phi_{b,j}(t_f) \rangle$ lies on the upper branch of replica 2. Define the normalized two-replica Wightman function in this χ -twisted representation by

$$G_{RR;\alpha\beta}^{(2),\eta;\chi\text{-tw},>}(t_1, t_2; t_f) = -\frac{i}{N_\eta} \sum_{j=1}^{N_\eta} \left\langle \eta_{R,j}^\alpha(t_{1,-}) \eta_{R,j}^\beta(t_{2,+}) \right\rangle_{\chi\text{-twisted}}. \quad (\text{S135})$$

Here the twisted expectation value is normalized by the unperturbed two-replica partition function, equivalently by $\mathcal{P}_\chi(t_f)$. With the branch and replica assignments above, Eq. (S134) becomes

$$F_2(t_b; t_f) = 2i G_{RR;12}^{(2),\eta;\chi\text{-tw},>}(t_b, t_b; t_f). \quad (\text{S136})$$

Signed fermionic swap and the untwisted bath frame. The replica gluing must retain the sign associated with a fermionic trace. For a fermionic operator O , the coherent-state representation of the trace is

$$\text{Tr}_\chi O = \int d\bar{\xi} d\xi e^{-\bar{\xi}\xi} \langle -\xi | O | \xi \rangle. \quad (\text{S137})$$

Applying this identity to $\text{Tr}_\chi[\tau_{\chi,j}^\dagger \tau_{\chi,j}]$ and inserting a coherent-state resolution of the identity between the two transition matrices gives

$$\begin{aligned} \text{Tr}_\chi [\tau_{\chi,j}^\dagger \tau_{\chi,j}] &= \int d\bar{\xi}_0 d\xi_0 d\bar{\xi}_1 d\xi_1 e^{-\bar{\xi}_0 \xi_0 - \bar{\xi}_1 \xi_1} \\ &\times \langle -\xi_0 | \tau_{\chi,j}^\dagger | \xi_1 \rangle \langle \xi_1 | \tau_{\chi,j} | \xi_0 \rangle. \end{aligned} \quad (\text{S138})$$

The endpoints of the two transition strips are consequently identified as

$$\chi^1(t_{f,+}) = -\chi^2(t_{f,-}), \quad \chi^2(t_{f,+}) = \chi^1(t_{f,-}). \quad (\text{S139})$$

Writing the two replica fields as a column vector, this becomes

$$\begin{pmatrix} \chi^1(t_{f,+}) \\ \chi^2(t_{f,+}) \end{pmatrix} = \mathbf{S} \begin{pmatrix} \chi^1(t_{f,-}) \\ \chi^2(t_{f,-}) \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}. \quad (\text{S140})$$

The minus sign in the first row originates from the anti-periodic trace in Eq. (S137). The signed swap obeys

$$\mathbf{S}^\dagger = \mathbf{S}^T = -\mathbf{S}, \quad \mathbf{S}^\dagger \mathbf{S} = \mathbf{1}, \quad \mathbf{S}^2 = -\mathbf{1}. \quad (\text{S141})$$

For later use, divide the folded contour into its upper and lower folds,

$$\mathcal{C}_u = \mathcal{C}_{\beta/4}^+ \cup \mathcal{C}^+, \quad \mathcal{C}_l = \mathcal{C}^- \cup \mathcal{C}_{\beta/4}^-. \quad (\text{S142})$$

Introduce the branch-frame rotation

$$\mathbf{U}(z) = \begin{cases} \mathbf{1}, & z \in \mathcal{C}_u, \\ \mathbf{S}, & z \in \mathcal{C}_l. \end{cases} \quad (\text{S143})$$

The symbol P is therefore reserved exclusively for the auxiliary fermionic reference mode introduced below.

Define the untwisted bath field by

$$\tilde{\chi}(z) = \mathbf{U}(z) \chi(z), \quad \chi(z) = \mathbf{U}^\dagger(z) \tilde{\chi}(z). \quad (\text{S144})$$

At the final-time endpoint,

$$\tilde{\chi}(t_{f,+}) = \chi(t_{f,+}), \quad \tilde{\chi}(t_{f,-}) = \mathbf{S} \chi(t_{f,-}),$$

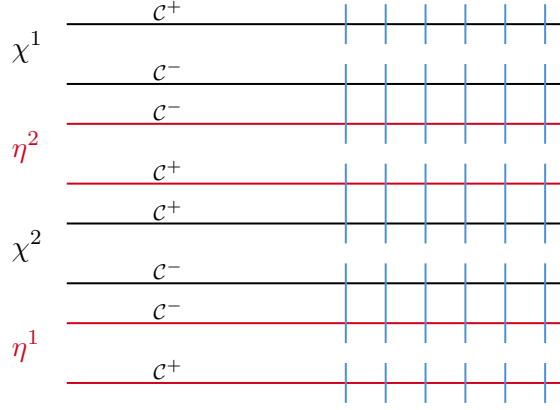


Figure S2: **From the η -twisted to the χ -twisted representation.** The reduced-state contraction $\text{Tr}_\eta(\sigma_{\eta,j}\rho_\eta)$ cyclically glues the η indices between replicas, while the χ indices close within each replica. Regrouping the same wave-function coefficients as in Eq. (S133) gives the identical contraction $\text{Tr}_\chi(\tau_{\chi,j}^\dagger\tau_{\chi,j})$, in which the η indices close within each transition strip and the signed replica twist is carried by χ . After the branch-dependent rotation in Eq. (S143), this equivalence is represented by the replica-diagonal pairings (χ^1, η^1) and (χ^2, η^2) on $\mathcal{C}^+$, and the crossed pairings (χ^1, η^2) and (χ^2, η^1) on $\mathcal{C}^-$. Red and black curves denote the η and χ contours, respectively, while the blue links indicate their mixed-interaction pairing. Here R, L copy indices are suppressed.

so Eq. (S140) implies the ordinary gluing condition

$$\tilde{\chi}(t_{f,+}) = \tilde{\chi}(t_{f,-}). \quad (\text{S145})$$

Thus the twist has been moved from the endpoint condition into the branch-dependent replica frame.

The two-replica bilocals in this frame are

$$\begin{aligned} (G_{ab}^\eta)_{\alpha\beta}(z_1, z_2) &= -\frac{i}{N_\eta} \sum_{j=1}^{N_\eta} \eta_{a,j}^\alpha(z_1) \eta_{b,j}^\beta(z_2), \\ (\tilde{G}_{ab}^\chi)_{\alpha\beta}(z_1, z_2) &= -\frac{i}{N_\chi} \sum_{k=1}^{N_\chi} \tilde{\chi}_{a,k}^\alpha(z_1) \tilde{\chi}_{b,k}^\beta(z_2). \end{aligned} \quad (\text{S146})$$

The physical χ bilocal in the twisted frame is therefore

$$G_{ab}^{\chi,\text{tw}}(z_1, z_2) = \mathbf{U}^\dagger(z_1) \tilde{G}_{ab}^\chi(z_1, z_2) \mathbf{U}(z_2). \quad (\text{S147})$$

We rotate the bath self-energy in the same way,

$$\Sigma_{ab}^{\chi,\text{tw}}(z_1, z_2) = \mathbf{U}^\dagger(z_1) \tilde{\Sigma}_{ab}^\chi(z_1, z_2) \mathbf{U}(z_2). \quad (\text{S148})$$

Since $\mathbf{U}$ is a real orthogonal signed-permutation matrix, the componentwise bilocal pairing is invariant:

$$\sum_{\alpha,\beta} (\Sigma_{ab}^{\chi,\text{tw}})_{\alpha\beta} (G_{ab}^{\chi,\text{tw}})_{\alpha\beta} = \sum_{\alpha,\beta} (\tilde{\Sigma}_{ab}^\chi)_{\alpha\beta} (\tilde{G}_{ab}^\chi)_{\alpha\beta}. \quad (\text{S149})$$

The kinetic and local left-right bilinear terms are also invariant under Eq. (S144). The pure χ interaction is invariant because left and right multiplication by $\mathbf{U}$ only permutes matrix entries and changes some signs, while p is even:

$$\sum_{\alpha,\beta} \left[2 \left(\mathbf{U}^\dagger(z_1) \tilde{G}_{ab}^\chi(z_1, z_2) \mathbf{U}(z_2) \right)_{\alpha\beta} \right]^p = \sum_{\alpha,\beta} \left[2 (\tilde{G}_{ab}^\chi)_{\alpha\beta}(z_1, z_2) \right]^p. \quad (\text{S150})$$

The rotation cannot be removed from the mixed interaction because the η bilocal remains in the original replica frame. The post-evaporation mixed combination is therefore

$$s(G_{ab}^\eta)_{\alpha\beta} + (1-s)(G_{ab}^{\chi,\text{tw}})_{\alpha\beta}. \quad (\text{S151})$$

Two-replica effective action and probe limit. The two-replica bare inverse propagator is the replica-diagonal lift of Eq. (S20),

$$(G_{0,2}^{-1})_{\alpha\beta;ab}(z_1, z_2) = \delta_{\alpha\beta} i\delta_{ab} \partial_{z_1} \delta_{\mathcal{C}}(z_1, z_2). \quad (\text{S152})$$

Using the normalization of the one-replica action (S19), the finite- s , finite- p action in the χ -twisted representation is

$$\begin{aligned} \mathcal{S}_{2\text{rep}} = & -s \frac{i}{2} \text{Tr} \log \left[-i \left(G_{0,2}^{-1} - \Sigma^\eta \right) \right] - (1-s) \frac{i}{2} \text{Tr} \log \left[-i \left(G_{0,2}^{-1} - \tilde{\Sigma}^\chi \right) \right] \\ & + \frac{i}{2} \int_{\mathcal{C}} dz_1 dz_2 \sum_{a,b} \sum_{\alpha,\beta=1}^2 \left[s (\Sigma_{ab}^\eta)_{\alpha\beta} (G_{ab}^\eta)_{\alpha\beta} + (1-s) (\tilde{\Sigma}_{ab}^\chi)_{\alpha\beta} (\tilde{G}_{ab}^\chi)_{\alpha\beta} \right] \\ & - \frac{i\mathcal{J}^2}{4p^2} \int_{\mathcal{C}} dz_1 dz_2 \sum_{a,b} \sum_{\alpha,\beta=1}^2 (-1)^{1+p\delta_{ab}/2} \left[l_\eta(z_1) l_\eta(z_2) [2(G_{ab}^\eta)_{\alpha\beta}]^p \right. \\ & \quad + l_\chi(z_1) l_\chi(z_2) [2(\tilde{G}_{ab}^\chi)_{\alpha\beta}]^p \\ & \quad + l_c(z_1) l_c(z_2) \left\{ \left[2 \left(s(G_{ab}^\eta)_{\alpha\beta} + (1-s)(G_{ab}^{\chi,\text{tw}})_{\alpha\beta} \right) \right]^p \right. \\ & \quad \quad \left. \left. - s^p [2(G_{ab}^\eta)_{\alpha\beta}]^p - (1-s)^p [2(\tilde{G}_{ab}^\chi)_{\alpha\beta}]^p \right] \right\} \\ & \quad \left. + \frac{1}{2p} \int_{\mathcal{C}} dz_1 dz_2 \delta_{\mathcal{C}}(z_1, z_2) \sum_{\alpha=1}^2 \left[s \mu_\eta(z_1) ((G_{LR}^\eta)_{\alpha\alpha} - (G_{RL}^\eta)_{\alpha\alpha}) \right. \right. \\ & \quad \left. \left. + (1-s) \mu_\chi(z_1) ((\tilde{G}_{LR}^\chi)_{\alpha\alpha} - (\tilde{G}_{RL}^\chi)_{\alpha\alpha}) \right] \right]. \quad (\text{S153}) \end{aligned}$$

All bilocals in Eq. (S153) are evaluated at (z_1, z_2) unless their arguments are shown explicitly. The trace in the first line includes the contour, the R, L indices, and the replica indices. The profiles l_η, l_χ, l_c and μ_η, μ_χ are precisely those defined in Eqs. (S3) and (S4); no additional bilinear-profile functions are introduced here.

For the two-replica calculation below, we work in the large- N probe, or large-bath, limit

$$N_\eta, N_\chi \longrightarrow \infty, \quad s \longrightarrow 0, \quad b = d = 1, \quad p \text{ fixed}, \quad p/2 \text{ even}. \quad (\text{S154})$$

At fixed p , $sp \rightarrow 0$ follows from $s \rightarrow 0$, and the χ saddle has no backreaction from the η sector. Throughout the remainder of this section, the two-replica equations are studied at finite p . An analytic treatment of the two-replica problem in the additional large- p limit is left for future work.

In the untwisted frame the bath saddle is replica diagonal:

$$(\tilde{G}_{ab}^\chi)_{\alpha\beta}(z_1, z_2) = \delta_{\alpha\beta} G_{ab}^\chi(z_1, z_2), \quad (\text{S155})$$

where G_{ab}^χ is the ordinary one-replica bath solution of Eqs. (S24) and (S21). Transforming back to the physical twisted frame gives

$$(G_{ab}^{\chi,\text{tw}})_{\alpha\beta}(z_1, z_2) = G_{ab}^\chi(z_1, z_2) \Pi_{\alpha\beta}(z_1, z_2), \quad \Pi(z_1, z_2) \equiv \mathbf{U}^\dagger(z_1) \mathbf{U}(z_2). \quad (\text{S156})$$

Explicitly,

$$\Pi(z_1, z_2) = \begin{cases} \mathbf{1}, & z_1, z_2 \in \mathcal{C}_u \text{ or } z_1, z_2 \in \mathcal{C}_l, \\ \mathbf{S}, & z_1 \in \mathcal{C}_u, \quad z_2 \in \mathcal{C}_l, \\ \mathbf{S}^\dagger = -\mathbf{S}, & z_1 \in \mathcal{C}_l, \quad z_2 \in \mathcal{C}_u. \end{cases} \quad (\text{S157})$$

Every entry of Π is 0 or ± 1 . Since $p-1$ is odd, the elementwise power appearing in the self-energy satisfies

$$[2G_{ab}^\chi(z_1, z_2) \Pi(z_1, z_2)]^{o(p-1)} = [2G_{ab}^\chi(z_1, z_2)]^{p-1} \Pi(z_1, z_2), \quad (\text{S158})$$

where the superscript $\circ(p-1)$ denotes an elementwise power in replica space.

For the sharp protocol, let $\mathcal{C}_<$ denote the Euclidean preparation segments together with the real-time portions before t_{ev} , and let $\mathcal{C}_>$ denote the real-time portions after t_{ev} . Varying Eq. (S153) and taking the limit (S154) gives

$$(\Sigma_{ab}^\eta)_{\alpha\beta}(z_1, z_2) = (-1)^{1+p\delta_{ab}/2} \frac{\mathcal{J}^2}{p} \begin{cases} a^2 [2(G_{ab}^\eta)_{\alpha\beta}(z_1, z_2)]^{p-1}, & z_1, z_2 \in \mathcal{C}_<, \\ [2G_{ab}^\chi(z_1, z_2)]^{p-1} \Pi_{\alpha\beta}(z_1, z_2), & z_1, z_2 \in \mathcal{C}_>, \\ 0, & z_1 \in \mathcal{C}_<, z_2 \in \mathcal{C}_> \text{ or vice versa} \end{cases} \\ - i\epsilon_{ab} \delta_{\alpha\beta} \frac{\mu_\eta(z_1)}{p} \delta_{\mathcal{C}}(z_1, z_2), \quad (\text{S159})$$

where ϵ_{ab} is defined in Eq. (S22). The pre-evaporation replica problem retains the nonlinear η self-energy, whereas after evaporation the η probe is driven by the known scalar bath solution multiplied by the signed replica kernel Π . There is no interaction self-energy when the two arguments lie on opposite sides of the evaporation surface. The probe calculation therefore requires no new nonlinear bath saddle.

Numerical two-replica equations. The scalar bath correlator G^χ is first obtained from the ordinary one-replica equations (S24) and (S21). The resulting bath solution is then held fixed while solving the two-replica η equation.

Discretize the contour by points z_i , ordered so that $i > j$ means $z_i >_{\mathcal{C}} z_j$. Each fold contains a Euclidean preparation segment of target length $\beta/4$ and a real-time segment ending at t_f . The implementation uses real-time spacing Δt and chooses the integer node counts as follows:

$$N_\beta = \text{round}\left(\frac{\beta}{4\Delta t}\right), \quad N_t = \text{round}\left(\frac{t_f}{\Delta t}\right), \quad N_{\text{fold}} = N_\beta + N_t, \quad N_{\mathcal{C}} = 2N_{\text{fold}}. \quad (\text{S160})$$

The plotted final times are grid aligned, so $N_t \Delta t = t_f$. The Euclidean spacing is chosen separately as $\Delta\tau = \beta/(4N_\beta)$, so that each preparation segment has length $N_\beta \Delta\tau = \beta/4$ and $\beta_{\text{grid}} = 4N_\beta \Delta\tau = \beta$, up to floating-point precision. For Fig. 4, $\beta = 5.7355465869$ and $\Delta t = 0.5$ give $N_\beta = 3$ and $\Delta\tau \simeq 0.4779622156$. The contour measure is represented by the diagonal weight matrix

$$W_{ij}^{ab} = \delta_{ab} \delta_{ij} w_i, \quad w_i = \begin{cases} -i\Delta\tau, & 0 \leq i < N_\beta, \\ +\Delta t, & N_\beta \leq i < N_{\text{fold}}, \\ -\Delta t, & N_{\text{fold}} \leq i < N_{\text{fold}} + N_t, \\ -i\Delta\tau, & N_{\text{fold}} + N_t \leq i < 2N_{\text{fold}}. \end{cases} \quad (\text{S161})$$

Using the composite index

$$\mathcal{A} = (\alpha, a, i), \quad \alpha = 1, 2, \quad a \in \{R, L\}, \quad i = 0, \dots, N_{\mathcal{C}} - 1,$$

the replica-diagonal free propagator and weight matrix are

$$(\mathbf{G}_{0,2})_{\alpha ai, \beta bj} = \delta_{\alpha\beta} (\mathbf{G}_0)_{ai, bj}, \quad (\text{S162}) \\ (\mathbf{W}_2)_{\alpha ai, \beta bj} = \delta_{\alpha\beta} \delta_{ab} \delta_{ij} w_i.$$

At the discrete level the signed twist is

$$\Pi_{ij} = \mathbf{U}^\dagger(z_i) \mathbf{U}(z_j). \quad (\text{S163})$$

Let T_i be the real time associated with z_i , with Euclidean preparation points assigned to the pre-evaporation region. Define the masks

$$M_{\text{I}}(i, j) = \mathbf{1}_{T_i \leq t_{\text{ev}}} \mathbf{1}_{T_j \leq t_{\text{ev}}}, \\ M_{\text{III}}(i, j) = \mathbf{1}_{T_i > t_{\text{ev}}} \mathbf{1}_{T_j > t_{\text{ev}}}, \quad (\text{S164}) \\ M_{\text{II}}(i, j) = 1 - M_{\text{I}}(i, j) - M_{\text{III}}(i, j).$$

Here $\mathbf{1}$ is an indicator function. A grid point exactly at t_{ev} belongs to the pre-evaporation region, so the masks form a partition also on the quench slice. The pre-evaporation self-energy is nonlinear in the unknown two-replica correlator:

$$\left(\Sigma_{ab}^{\eta, \text{I}}\right)_{\alpha\beta;ij} = M_{\text{I}}(i, j)(-1)^{1+p\delta_{ab}/2} \frac{a^2 \mathcal{J}^2}{p} [2(G_{ab}^{\eta})_{\alpha\beta;ij}]^{p-1}. \quad (\text{S165})$$

The post-evaporation self-energy is fixed by the bath:

$$\left(\Sigma_{ab}^{\eta, \text{III}}\right)_{\alpha\beta;ij} = M_{\text{III}}(i, j)(-1)^{1+p\delta_{ab}/2} \frac{\mathcal{J}^2}{p} [2G_{ab;ij}^{\chi}]^{p-1} (\Pi_{ij})_{\alpha\beta}. \quad (\text{S166})$$

There is no interaction contribution in region II. A local η bilinear, if retained, is represented by

$$(\mathbf{V}_{\eta})_{\alpha\alpha i, \beta\beta j} = -i\epsilon_{ab} \delta_{\alpha\beta} \delta_{ij} \frac{\mu_{\eta}(z_i)}{p}. \quad (\text{S167})$$

For the protocol used in the numerical results, $\mu_{\eta} = 0$, so $\mathbf{V}_{\eta} = 0$.

The discrete two-replica Dyson equation is

$$\mathbf{G}_{\eta} = \mathbf{G}_{0,2} + \mathbf{G}_{0,2} \left[\mathbf{W}_2 \mathbf{V}_{\eta} + \mathbf{W}_2 (\Sigma^{\eta, \text{I}}[\mathbf{G}_{\eta}] + \Sigma_{\chi}^{\eta, \text{III}}) \mathbf{W}_2 \right] \mathbf{G}_{\eta}. \quad (\text{S168})$$

The local vertex carries one contour weight, whereas a nonlocal bilocal self-energy carries one weight for each contour argument.

Because the post-evaporation bath source is fixed, define

$$\mathbf{A}_{\chi} = \mathbf{1} - \mathbf{G}_{0,2} [\mathbf{W}_2 \mathbf{V}_{\eta} + \mathbf{W}_2 \Sigma_{\chi}^{\eta, \text{III}} \mathbf{W}_2]. \quad (\text{S169})$$

The remaining nonlinear residual is

$$R_{\eta}(\mathbf{G}_{\eta}) = \mathbf{A}_{\chi} \mathbf{G}_{\eta} - \mathbf{G}_{0,2} - \mathbf{G}_{0,2} \mathbf{W}_2 \Sigma^{\eta, \text{I}}[\mathbf{G}_{\eta}] \mathbf{W}_2 \mathbf{G}_{\eta}. \quad (\text{S170})$$

We solve $R_{\eta} = 0$ using a damped Newton–Krylov method. The Jacobian-vector product is

$$J_{\eta}[\mathbf{G}_{\eta}] \mathbf{X} = \mathbf{A}_{\chi} \mathbf{X} - \mathbf{G}_{0,2} \mathbf{W}_2 \delta \Sigma^{\eta, \text{I}}[\mathbf{X}] \mathbf{W}_2 \mathbf{G}_{\eta} - \mathbf{G}_{0,2} \mathbf{W}_2 \Sigma^{\eta, \text{I}}[\mathbf{G}_{\eta}] \mathbf{W}_2 \mathbf{X}, \quad (\text{S171})$$

where

$$\left(\delta \Sigma_{ab}^{\eta, \text{I}}[\mathbf{X}]\right)_{\alpha\beta;ij} = M_{\text{I}}(i, j)(-1)^{1+p\delta_{ab}/2} \frac{a^2 \mathcal{J}^2}{p} (p-1) 2 [2(G_{ab}^{\eta})_{\alpha\beta;ij}]^{p-2} (X_{ab})_{\alpha\beta;ij}. \quad (\text{S172})$$

At each Newton step, we use the generalized minimal residual method (GMRES) to solve the linearized equation $J_{\eta}[\mathbf{G}_{\eta}] \delta \mathbf{G}_{\eta} = -R_{\eta}(\mathbf{G}_{\eta})$. The solver requires only the Jacobian-vector product in Eq. (S171), so the full Jacobian is never constructed. Starting from the linearized residual $\mathbf{r}$, GMRES builds the Krylov subspace $\text{span}\{\mathbf{r}, J_{\eta} \mathbf{r}, \dots, J_{\eta}^{m-1} \mathbf{r}\}$ and chooses the correction that minimizes the residual norm within this subspace. In practice, $\mathbf{A}_{\chi}^{-1}$ is used as a preconditioner: it approximately removes the stiff linear contribution from the fixed bath, improving the conditioning of the linear system and accelerating GMRES without changing the desired solution.

If $i_{-}(t_b)$ and $i_{+}(t_b)$ are the lower- and upper-branch indices corresponding to t_b , the normalized overlap is read off as

$$F_2(t_b; t_f) = 2i (\mathbf{G}_{\eta})_{(1, R, i_{-}(t_b)), (2, R, i_{+}(t_b))}. \quad (\text{S173})$$

Rényi-2 mutual information. For compactness, define the branch-resolved contractions

$$C_{\alpha\beta}^{\lambda_1 \lambda_2}(t_b; t_f) \equiv 2i G_{RR; \alpha\beta}^{(2), \eta; \chi\text{-tw}}(t_b, \lambda_1, t_b, \lambda_2; t_f), \quad \lambda_1, \lambda_2 \in \{+, -\}. \quad (\text{S174})$$

Thus

$$F_2(t_b; t_f) = C_{12}^{-+}(t_b; t_f). \quad (\text{S175})$$

The second contraction required below is the fixed-flavor perturbed purity, averaged over the flavor index:

$$F_4(t_b; t_f) = \frac{1}{N_\eta} \sum_{j=1}^{N_\eta} \frac{\text{Tr}_\eta [\sigma_{\eta,j}(t_b; t_f)^2]}{\mathcal{P}_\eta(t_f)}. \quad (\text{S176})$$

At leading order in large N_η , the corresponding four-point function factorizes into two-replica two-point functions. Keeping the fermionic Wick signs gives

$$F_4(t_b; t_f) = C_{11}^{-+}(t_b; t_f)C_{22}^{-+}(t_b; t_f) - C_{12}^{-+}(t_b; t_f)C_{21}^{-+}(t_b; t_f) - C_{12}^{--}(t_b; t_f)C_{12}^{++}(t_b; t_f). \quad (\text{S177})$$

The last term pairs the two lower-branch insertions with each other and the two upper-branch insertions with each other. Thus no independent four-point saddle is required.

The corresponding χ -side cross contraction is

$$\begin{aligned} K_2(t_b; t_f) &= \frac{1}{N_\eta} \sum_{j=1}^{N_\eta} \frac{\text{Tr}_\chi [\sigma_{\chi,j}(t_b; t_f)\rho_\chi(t_f)]}{\mathcal{P}_\chi(t_f)} \\ &= C_{11}^{-+}(t_b; t_f) = 2i G_{RR;11}^{(2),\eta;\chi\text{-tw},>}(t_b, t_b; t_f). \end{aligned} \quad (\text{S178})$$

To retain the coherence of the infalling excitation, introduce an auxiliary two-dimensional reference system P and purify the choice of whether or not the simple fermion was inserted:

$$|\Omega_j(t_f)\rangle_{P\eta\chi} = \frac{1}{\sqrt{2}} [|0\rangle_P |\Phi(t_f)\rangle_{\eta\chi} + |1\rangle_P |\Phi_{b,j}(t_f)\rangle_{\eta\chi}]. \quad (\text{S179})$$

Since $\eta_j^R(t_b)$ is fermion odd, the states $|\Phi(t_f)\rangle$ and $|\Phi_{b,j}(t_f)\rangle$ have opposite fermion parity and are therefore orthogonal. It follows that the reference system is maximally mixed:

$$\rho_P = \frac{1}{2} (|0\rangle\langle 0| + |1\rangle\langle 1|), \quad S_P^{(2)} = -\log \text{Tr} \rho_P^2 = \log 2. \quad (\text{S180})$$

The Rényi-2 mutual-information diagnostic between P and η is

$$I^{(2)}(P : \eta) = S_P^{(2)} + S_\eta^{(2)} - S_{P\eta}^{(2)}. \quad (\text{S181})$$

The full $P\eta\chi$ state is pure, so $S_{P\eta}^{(2)} = S_\chi^{(2)}$. Therefore

$$\begin{aligned} e^{-I^{(2)}(P:\eta)} &= \frac{\text{Tr}_\eta [(\rho_\eta + \sigma_{\eta,j})^2]}{2 \text{Tr}_\chi [(\rho_\chi + \sigma_{\chi,j})^2]} \\ &= \frac{1}{2} \frac{1 + 2F_2 + F_4}{1 + 2K_2 + F_4}. \end{aligned} \quad (\text{S182})$$

The first line is precisely the density-matrix expression quoted in Eq. (14). The second line uses flavor symmetry at the leading disorder-averaged large- N saddle, where the fixed-flavor contractions equal F_2, F_4, K_2 ; it is not an identity between averages of ratios in an arbitrary finite- N realization. In the ideal limit in which P and the retained η information are maximally entangled, this diagnostic equals $2 \log 2$.

Projection onto the reconstructed bulk mode. Equations (S136), (S177), and (S178) were written first for a simple boundary insertion. The main-text result instead uses the reconstructed infalling mode $\psi_{R,f}(u, t_{\text{max}})$. According to Eq. (26), this operator is a linear combination of boundary operators over the reconstruction interval.

Let $\mathbf{K}_f(u, t_{\text{max}})$ denote the corresponding discretized reconstruction row, including the boundary-copy components used in the reconstruction. For an insertion on the ket or upper branch and on the bra or lower branch, respectively, use

$$\mathbf{K}_f^{(+)} = \mathbf{K}_f, \quad \mathbf{K}_f^{(-)} = \mathbf{K}_f^*. \quad (\text{S183})$$

If $\mathbf{C}_{\alpha\beta}^{\lambda_1\lambda_2}$ denotes the branch-resolved two-replica correlator matrix assembled from the RR, RL, LR, LL boundary blocks, define its bulk projection by

$$C_{\alpha\beta}^{\lambda_1\lambda_2}[\mathbf{K}_f] = \mathbf{K}_f^{(\lambda_1)} \mathbf{C}_{\alpha\beta}^{\lambda_1\lambda_2} \left(\mathbf{K}_f^{(\lambda_2)} \right)^T. \quad (\text{S184})$$

A boundary insertion at t_b is recovered by taking $\mathbf{K}_f$ to be the corresponding delta-function row. For the reconstructed bulk mode, the quantities entering the mutual information are

$$\begin{aligned} F_2[\mathbf{K}_f] &= C_{12}^{-+}[\mathbf{K}_f], \\ K_2[\mathbf{K}_f] &= C_{11}^{-+}[\mathbf{K}_f], \\ F_4[\mathbf{K}_f] &= C_{11}^{-+}[\mathbf{K}_f] C_{22}^{-+}[\mathbf{K}_f] - C_{12}^{-+}[\mathbf{K}_f] C_{21}^{-+}[\mathbf{K}_f] - C_{12}^{-+}[\mathbf{K}_f] C_{12}^{++}[\mathbf{K}_f]. \end{aligned} \quad (\text{S185})$$

Substitution of these kernel-projected contractions into Eq. (S182) gives the bulk-mode Rényi-2 mutual information shown in Fig. 4.

On-shell action and second Rényi entropy. The unperturbed two-replica saddles also determine the second Rényi entropy of the full η sector. Let the statistical on-shell action be

$$\mathcal{I} = -i \mathcal{S}_{2\text{rep}, \text{on-shell}}, \quad (\text{S186})$$

where the contour action includes both Euclidean preparation segments and both real-time branches. The contour weights in Eq. (S161) already include all branch orientations.

The purity is the ratio of the χ -twisted partition function to the disconnected two-replica partition function:

$$\text{Tr } \rho_\eta(t_f)^2 = \frac{Z_{\chi\text{-tw}}(t_f)}{Z_{\text{disc}}(t_f)} \sim \exp[-(N_\eta + N_\chi)(\mathcal{I}_{\chi\text{-tw}} - \mathcal{I}_{\text{disc}})]. \quad (\text{S187})$$

In the probe limit the statistical action has the expansion

$$\mathcal{I} = \mathcal{I}_\chi + s i_\eta + O(s^2), \quad s = \frac{N_\eta}{N_\eta + N_\chi}. \quad (\text{S188})$$

The bath saddle is common to the twisted and disconnected solutions at this order, so its contribution cancels. Define

$$\Delta i_\eta(t_f) = i_{\eta, \chi\text{-tw}}(t_f) - i_{\eta, \text{disc}}(t_f). \quad (\text{S189})$$

The entropy normalization is then

$$\frac{S_\eta^{(2)}(t_f)}{N_\eta} = \text{Re } \Delta i_\eta(t_f), \quad \frac{S_\eta^{(2)}(t_f)}{N_\eta + N_\chi} = s \text{Re } \Delta i_\eta(t_f). \quad (\text{S190})$$

Here $S_\eta^{(2)}(t_f) = -\log \text{Tr } \rho_\eta(t_f)^2$. The imaginary part of Δi_η is a numerical convergence diagnostic and should vanish for a converged physical saddle.

The doubled η system contains $2N_\eta$ Majorana fermions and has Hilbert-space dimension 2^{N_η} . Consequently,

$$0 \leq \frac{S_\eta^{(2)}(t_f)}{N_\eta} = \text{Re } \Delta i_\eta(t_f) \leq \log 2. \quad (\text{S191})$$

For the χ -twisted saddle, the post-evaporation source is the one in Eq. (S166). The disconnected saddle uses the same scalar bath correlator but no replica mixing:

$$\left(\Sigma_{\text{disc}, \text{ab}}^{\eta, \text{III}} \right)_{\alpha\beta; ij} = M_{\text{III}}(i, j) (-1)^{1+p\delta_{ab}/2} \frac{\mathcal{J}^2}{p} \left[2G_{ab; ij}^\chi \right]^{p-1} \delta_{\alpha\beta}. \quad (\text{S192})$$

For the convention used in the plots, $p/2$ is even, so $(-1)^{1+p\delta_{ab}/2} = -1$ in every R, L block. We also use $\mathbf{V}_\eta = 0$.

The free determinant is identical for the twisted and disconnected saddles. Writing

$$G_{0,2}^{-1} - \Sigma^\eta = G_{0,2}^{-1} (\mathbf{1} - G_{0,2} \Sigma^\eta)$$

makes this cancellation explicit. The UV-subtracted discrete log-determinant contribution is

$$i_{\eta,\log} = -\frac{1}{2} \log \det [\mathbf{1} - \mathbf{G}_{0,2} \mathbf{W}_2 (\Sigma^{\eta,\text{I}} + \Sigma^{\eta,\text{III}}) \mathbf{W}_2]. \quad (\text{S193})$$

Both contour arguments of a nonlocal self-energy carry a quadrature weight.

At leading order in the probe expansion, the explicit region-III ΣG contribution cancels the region-III mixed potential. The post-evaporation self-energy nevertheless remains in the determinant and in the saddle equation. The surviving pre-evaporation interaction contribution is

$$i_{\eta,\text{I}} = \frac{a^2 \mathcal{J}^2}{4p^2} \sum_{\alpha,\beta=1}^2 \sum_{a,b=R,L} \sum_{i,j} w_i w_j M_{\text{I}}(i,j) (-1)^{1+p\delta_{ab}/2} [2(G_{ab}^\eta)_{\alpha\beta;ij}]^p. \quad (\text{S194})$$

The corresponding constraint contribution is

$$i_{\Sigma_{\text{I}}G} = \frac{1}{2} \sum_{\alpha,\beta=1}^2 \sum_{a,b=R,L} \sum_{i,j} w_i w_j (\Sigma_{ab}^{\eta,\text{I}})_{\alpha\beta;ij} (G_{ab}^\eta)_{\alpha\beta;ij}. \quad (\text{S195})$$

This is an elementwise bilocal contraction; it should not be replaced by an ordinary matrix trace, which reverses one of the matrix indices.

The η -normalized probe action evaluated on either saddle is therefore

$$i_\eta = i_{\eta,\log} + i_{\Sigma_{\text{I}}G} - i_{\eta,\text{I}}. \quad (\text{S196})$$

For each t_f , the required action difference is

$$\Delta i_\eta(t_f) = i_\eta [G_{\chi\text{-tw}}^\eta; \Sigma_{\chi\text{-tw}}^{\eta,\text{III}}] - i_\eta [G_{\text{disc}}^\eta; \Sigma_{\text{disc}}^{\eta,\text{III}}]. \quad (\text{S197})$$

Finally, at fixed contour extent and fixed discretization, $\Sigma^\eta = O(p^{-1})$ gives the formal large- p expansion of the relative determinant (the series converges when the spectral radius of $\mathbf{A}$ is less than one):

$$-\frac{1}{2} \log \det(\mathbf{1} - \mathbf{A}) = \frac{1}{2} \text{Tr } \mathbf{A} + \frac{1}{4} \text{Tr } \mathbf{A}^2 + \dots, \quad \mathbf{A} = \mathbf{G}_{0,2} \mathbf{W}_2 \Sigma^\eta \mathbf{W}_2. \quad (\text{S198})$$

Accordingly,

$$\Delta i_\eta(t_f) = \frac{c_1(t_f)}{p} + \frac{c_2(t_f)}{p^2} + O(p^{-3}), \quad (\text{S199})$$

unless an additional symmetry removes the leading coefficient. At fixed final time, this gives the large- p scaling $S_\eta^{(2)} = O(N_\eta/p)$ quoted in the main text. The numerical consistency checks are

$$\text{Re } \Delta i_\eta \geq 0, \quad \text{Im } \Delta i_\eta \simeq 0, \quad \Delta i_\eta(t_f \leq t_{\text{ev}}) = 0, \quad \text{Re } \Delta i_\eta \leq \log 2. \quad (\text{S200})$$

S6. Large- p conventions, Lensky-Qi parametrization, and the left-right bound

Reflection symmetry and the large- p ansatz. We first collect the symmetry, normalization, and branch conventions used in Secs. S2-S5. For either species $\psi \in \{\eta, \chi\}$, and for $p/2$ even, the doubled Hamiltonian is invariant under

$$\mathbf{R}[\psi_R] = -\psi_L, \quad \mathbf{R}[\psi_L] = \psi_R. \quad (\text{S201})$$

The bilinear $\psi_L \psi_R$ is invariant under the same transformation.

For real times, define

$$G_{ab}^{\psi,>}(t_1, t_2) = -i \langle \psi_a(t_1) \psi_b(t_2) \rangle, \quad a, b \in \{R, L\}. \quad (\text{S202})$$

Hermiticity and reflection symmetry imply

$$\begin{aligned} \left[G_{ab}^{\psi,>}(t_1, t_2) \right]^* &= -G_{ba}^{\psi,>}(t_2, t_1), \\ G_{LL}^{\psi,>}(t_1, t_2) &= G_{RR}^{\psi,>}(t_1, t_2), \quad G_{LR}^{\psi,>}(t_1, t_2) = -G_{RL}^{\psi,>}(t_1, t_2). \end{aligned} \quad (\text{S203})$$

Consequently,

$$\begin{aligned} G_{RR}^{\psi,>}(t_1, t_2) &= - \left[G_{RR}^{\psi,>}(t_2, t_1) \right]^*, \\ G_{RL}^{\psi,>}(t_1, t_2) &= \left[G_{RL}^{\psi,>}(t_2, t_1) \right]^*. \end{aligned} \quad (\text{S204})$$

For complex contour times, Hermiticity becomes

$$\left[G_{ab}^{\psi,>}(z_1, z_2) \right]^* = -G_{ba}^{\psi,>}(z_2^*, z_1^*). \quad (\text{S205})$$

The normalization and form of the large- p ansatz can be motivated from the free bilinear problem

$$H_{\text{bil}} = i\mu\psi_L\psi_R. \quad (\text{S206})$$

Here μ is the physical single-pair gap; in the large- p scaling of the model it is replaced by μ_ψ/p . With

$$c = \frac{\psi_L + i\psi_R}{\sqrt{2}}, \quad c^\dagger = \frac{\psi_L - i\psi_R}{\sqrt{2}}, \quad (\text{S207})$$

one has

$$H_{\text{bil}} = \mu \left(c^\dagger c - \frac{1}{2} \right). \quad (\text{S208})$$

In a thermal state $\rho = e^{-\beta H_{\text{bil}}}/Z_\beta$,

$$G_{RR}^>(t, 0) = -\frac{i}{2} \frac{\cosh \left[\frac{\mu}{2}(\beta - 2it) \right]}{\cosh \left(\frac{\beta\mu}{2} \right)} \xrightarrow{\beta\mu \rightarrow \infty} -\frac{i}{2} e^{-i\mu t}, \quad (\text{S209})$$

and

$$G_{RL}^>(t, 0) = -\frac{1}{2} \frac{\sinh \left[\frac{\mu}{2}(\beta - 2it) \right]}{\cosh \left(\frac{\beta\mu}{2} \right)} \xrightarrow{\beta\mu \rightarrow \infty} -\frac{1}{2} e^{-i\mu t}. \quad (\text{S210})$$

This motivates the large- p parametrization

$$\begin{aligned} G_{RR}^{\psi,>}(t_1, t_2) &= -\frac{i}{2} e^{g_R^\psi(t_1, t_2)/p} = G_{LL}^{\psi,>}(t_1, t_2), \\ G_{RL}^{\psi,>}(t_1, t_2) &= -\frac{1}{2} e^{g_L^\psi(t_1, t_2)/p} = -G_{LR}^{\psi,>}(t_1, t_2). \end{aligned} \quad (\text{S211})$$

Thus the subscript R on g_R^ψ denotes the diagonal RR/LL channel, whereas L on g_L^ψ denotes the off-diagonal RL/LR channel.

We choose the continuous logarithmic branch for which $g_{R,L}^\psi = O(1)$ as $p \rightarrow \infty$. Equations (S204) and (S211) then give

$$g_X^\psi(t_1, t_2) = \left[g_X^\psi(t_2, t_1) \right]^*, \quad X = R, L. \quad (\text{S212})$$

On the complex contour,

$$\left[g_X^\psi(z_1, z_2) \right]^* = g_X^\psi(z_2^*, z_1^*), \quad X = R, L. \quad (\text{S213})$$

In particular,

$$g_R^\psi(t, t) = 0, \quad g_L^\psi(t, t) \in \mathbb{R}. \quad (\text{S214})$$

These relations fix the conjugation and branch conventions used in the matching formulas of Sec. S3.

Lensky-Qi parametrization. Consider a real-time region in which the correlators satisfy the homogeneous Liouville equations with effective interaction scale $\mathcal{J}_{\text{eff}}$,

$$\partial_{t_1} \partial_{t_2} g_X(t_1, t_2) = 2\sigma_X \mathcal{J}_{\text{eff}}^2 e^{g_X(t_1, t_2)}, \quad \sigma_R = 1, \quad \sigma_L = -1. \quad (\text{S215})$$

For the bath, $\mathcal{J}_{\text{eff}} = \mathcal{J}$, whereas for the pre-evaporation η solution, $\mathcal{J}_{\text{eff}} = a\mathcal{J}$.

Following Lensky and Qi [2], the physical solution selected by the thermal gluing conditions (S40) and the equal-time data (S41) and (S42) may be described by a single complex function $y(t)$:

$$\begin{aligned} e^{g_R(t_1, t_2)} &= -\frac{y'(t_1)y'(t_2)^*}{\sin^2(\mathcal{J}_{\text{eff}}[y(t_1) - y(t_2)^*])}, \\ e^{g_L(t_1, t_2)} &= \frac{y'(t_1)y'(t_2)^*}{\cos^2(\mathcal{J}_{\text{eff}}[y(t_1) - y(t_2)^*])}. \end{aligned} \quad (\text{S216})$$

The variable denoted by y here is denoted by ψ in Ref. [2]; we use y to avoid confusion with the species label $\psi \in \{\eta, \chi\}$. Its dimensions are

$$[y] = [t] = [\mathcal{J}_{\text{eff}}]^{-1}. \quad (\text{S217})$$

We choose the physical lower-half-plane branch

$$\text{Im } y(t) < 0. \quad (\text{S218})$$

The equal-time normalization $g_R(t, t) = 0$ then requires

$$|y'(t)| = \sinh(2\mathcal{J}_{\text{eff}}|\text{Im } y(t)|). \quad (\text{S219})$$

It is useful to introduce real variables $\phi(t)$ and $p(t)$ by

$$y'(t) = |y'(t)|e^{ip(t)} = \frac{e^{ip(t)}}{\sqrt{e^{2\phi(t)} - 1}}. \quad (\text{S220})$$

Here $p(t)$ is the Lensky-Qi canonical phase conjugate to $\phi(t)$; it should not be confused with the interaction order p . The equal-time left-right correlator is

$$g_L(t, t) = -2\phi(t). \quad (\text{S221})$$

For a real bilinear coupling $\mu_{\text{eff}}(t)$, the variables (ϕ, p) evolve according to the classical Hamiltonian

$$\mathcal{H}_Q(\phi, p; t) = -2\mathcal{J}_{\text{eff}}\sqrt{1 - e^{-2\phi}}\cos p + \mu_{\text{eff}}(t)\phi, \quad \{\phi, p\}_{\text{P.B.}} = 1. \quad (\text{S222})$$

The corresponding Hamilton equations are

$$\begin{aligned} \dot{\phi} &= 2\mathcal{J}_{\text{eff}}\sqrt{1 - e^{-2\phi}}\sin p, \\ \dot{p} &= -\mu_{\text{eff}}(t) + 2\mathcal{J}_{\text{eff}}\frac{e^{-2\phi}}{\sqrt{1 - e^{-2\phi}}}\cos p. \end{aligned} \quad (\text{S223})$$

Lensky-Qi trajectories used in Secs. S3 and S4. For the pre-evaporation η segment, $\mathcal{J}_{\text{eff}} = a\mathcal{J}$ and $\mu_{\text{eff}} = 0$. In terms of the thermal angle ϑ defined in Eq. (S51), the corresponding trajectory is

$$\begin{aligned} \phi_\eta(t) &= \log \left[\frac{\cosh(2a\mathcal{J}t \sin \vartheta)}{\sin \vartheta} \right], \\ p_\eta(t) &= \tan^{-1} [\tan \vartheta \tanh(2a\mathcal{J}t \sin \vartheta)], \\ y_\eta(t) &= \frac{1}{a\mathcal{J}} \tan^{-1} \tanh \left(a\mathcal{J}t \sin \vartheta - \frac{i\vartheta}{2} \right). \end{aligned} \quad (\text{S224})$$

The inverse functions and logarithms in this expression are taken on their continuous branches. Substitution into Eq. (S216) reproduces the pre-evaporation correlators in Eq. (S50).

For the stationary traversable-wormhole bath, $\mathcal{J}_{\text{eff}} = \mathcal{J}$ and $\mu_{\text{eff}} = \mu_0$. It is the fixed point

$$\phi_\chi(t) = \phi_G, \quad p_\chi(t) = 0, \quad (\text{S225})$$

where ϕ_G and V_G are defined in Eq. (S48). A convenient representative of the corresponding y -trajectory is

$$y_\chi(t) = V_G(t - t_\star) - \frac{i}{2\mathcal{J}} \tanh^{-1}(r_G), \quad V_G = \frac{r_G}{\sqrt{1 - r_G^2}}, \quad (\text{S226})$$

where $t_\star$ is an arbitrary time origin. Substitution into Eq. (S216) reproduces the stationary bath correlators in Eq. (S49).

Disk-coordinate proof of the post-evaporation bound. We now prove Eq. (S67) under the assumptions of Eq. (S66), with h_χ defined in Eq. (S65). In this paragraph $g_{R,L} = g_{R,L}^\chi$, and therefore $\mathcal{J}_{\text{eff}} = \mathcal{J}$.

Introduce the disk coordinate

$$w(t) = e^{-2i\mathcal{J}y(t)}. \quad (\text{S227})$$

Because $\text{Im } y(t) < 0$,

$$|w(t)| = e^{2\mathcal{J} \text{Im } y(t)} < 1. \quad (\text{S228})$$

Equation (S219) becomes

$$|y'(t)| = \frac{1 - |w(t)|^2}{2|w(t)|}. \quad (\text{S229})$$

For $w_a = w(t_a)$ and $w_b = w(t_b)$,

$$\begin{aligned} |\sin(\mathcal{J}[y(t_a) - y(t_b)^*])|^2 &= \frac{|1 - w_a w_b^*|^2}{4|w_a||w_b|}, \\ |\cos(\mathcal{J}[y(t_a) - y(t_b)^*])|^2 &= \frac{|1 + w_a w_b^*|^2}{4|w_a||w_b|}. \end{aligned} \quad (\text{S230})$$

Define the disk invariant

$$\mathcal{P}(u, v) \equiv \frac{(1 - |u|^2)(1 - |v|^2)}{|1 - uv^*|^2}. \quad (\text{S231})$$

It is invariant under simultaneous automorphisms of the unit disk. Equations (S216), (S229), and (S230) give

$$\left| e^{g_R^\chi(t_a, t_b)} \right| = \mathcal{P}(w_a, w_b), \quad \left| e^{g_L^\chi(t_a, t_b)} \right| = \mathcal{P}(w_a, -w_b). \quad (\text{S232})$$

Let

$$w_i = w(t_i), \quad w_{\text{ev}} = w(t_{\text{ev}}). \quad (\text{S233})$$

Substitution into Eq. (S65) gives

$$h_\chi = \mathcal{P}(w_1, -w_2) \mathcal{P}(w_{\text{ev}}, -w_{\text{ev}}) \frac{\mathcal{P}(w_1, w_{\text{ev}}) \mathcal{P}(w_2, w_{\text{ev}})}{\mathcal{P}(w_1, -w_{\text{ev}}) \mathcal{P}(w_2, -w_{\text{ev}})}. \quad (\text{S234})$$

Here and below the arguments $(t_1, t_2; t_{\text{ev}})$ of h_χ are suppressed.

Apply the disk automorphism

$$z(w) = \frac{w - w_{\text{ev}}}{1 - w_{\text{ev}}^* w}. \quad (\text{S235})$$

It obeys

$$z(w_{\text{ev}}) = 0, \quad z(-w_{\text{ev}}) = -v, \quad z(-w) = -\frac{z(w) + v}{1 + v^* z(w)}, \quad \text{where } v = \frac{2w_{\text{ev}}}{1 + |w_{\text{ev}}|^2}. \quad (\text{S236})$$

Because $|w_{\text{ev}}| < 1$, one has $|v| < 1$.

Writing

$$z_j = z(w_j), \quad z'_j = z(-w_j), \quad (\text{S237})$$

the invariance of $\mathcal{P}$ reduces Eq. (S234) to

$$h_\chi = \frac{\mathcal{P}(z_1, 0)\mathcal{P}(z_2, 0)}{\mathcal{P}(z'_1, 0)\mathcal{P}(z'_2, 0)} \mathcal{P}(z_1, z'_2) \mathcal{P}(v, 0). \quad (\text{S238})$$

Using Eq. (S236) explicitly gives

$$h_\chi = \frac{(1 - |z_1|^2)(1 - |z_2|^2)|1 + z_1 v^*|^2 |1 + z_2 v^*|^2}{|1 + z_1 v^* + z_2^* v + z_1 z_2^*|^2}. \quad (\text{S239})$$

Remove the phase of v by writing

$$v = \rho e^{i\alpha_0}, \quad \rho = |v| < 1, \quad \tau_j = e^{-i\alpha_0} z_j, \quad (\text{S240})$$

with an arbitrary choice of α_0 if $v = 0$. Then

$$h_\chi = \frac{(1 - |\tau_1|^2)(1 - |\tau_2|^2)|1 + \rho\tau_1|^2 |1 + \rho\tau_2|^2}{|1 + \rho\tau_1 + \rho\tau_2^* + \tau_1\tau_2^*|^2}, \quad |\tau_1|, |\tau_2| < 1. \quad (\text{S241})$$

Notice that h_χ is already a nonnegative modulus; the ratio in Eq. (S241) is h_χ , not h_χ^2 .

To bound this expression, define

$$\kappa = \frac{\rho + \tau_1}{1 + \rho\tau_1}. \quad (\text{S242})$$

Since ρ is real,

$$1 - |\kappa|^2 = \frac{(1 - \rho^2)(1 - |\tau_1|^2)}{|1 + \rho\tau_1|^2} > 0, \quad (\text{S243})$$

so κ lies in the open unit disk. Moreover,

$$1 + \rho\tau_1 + \rho\tau_2^* + \tau_1\tau_2^* = (1 + \rho\tau_1)(1 + \kappa\tau_2^*). \quad (\text{S244})$$

Consequently,

$$h_\chi = (1 - |\tau_2|^2) |F(\tau_2)|^2, \quad (\text{S245})$$

where

$$F(\lambda) = \sqrt{1 - |\tau_1|^2} \frac{1 + \rho\lambda}{1 + \kappa^*\lambda}. \quad (\text{S246})$$

Because $|\kappa| < 1$, F is holomorphic on the unit disk. Its Taylor series is

$$F(\lambda) = \sqrt{1 - |\tau_1|^2} \left[1 + (\rho - \kappa^*) \sum_{n=1}^{\infty} (-\kappa^*)^{n-1} \lambda^n \right]. \quad (\text{S247})$$

Writing

$$F(\lambda) = \sum_{n=0}^{\infty} d_n \lambda^n, \quad (\text{S248})$$

the coefficients are

$$d_0 = \sqrt{1 - |\tau_1|^2}, \quad d_n = \sqrt{1 - |\tau_1|^2} (\rho - \kappa^*) (-\kappa^*)^{n-1}, \quad n \geq 1. \quad (\text{S249})$$

The associated Hardy norm is therefore finite and equals

$$\begin{aligned} \|F\|_{H^2}^2 &\equiv \sum_{n=0}^{\infty} |d_n|^2 \\ &= (1 - |\tau_1|^2) \left[1 + \frac{|\rho - \kappa^*|^2}{1 - |\kappa|^2} \right]. \end{aligned} \quad (\text{S250})$$

Since ρ is real,

$$\rho - \kappa = -\frac{(1 - \rho^2)\tau_1}{1 + \rho\tau_1}, \quad |\rho - \kappa^*| = |\rho - \kappa|. \quad (\text{S251})$$

Combining Eqs. (S243) and (S251) gives

$$\|F\|_{H^2}^2 = 1 - \rho^2|\tau_1|^2 \leq 1. \quad (\text{S252})$$

Because $F \in H^2$, Cauchy–Schwarz applied to its Taylor series gives

$$\begin{aligned} |F(\tau_2)|^2 &= \left| \sum_{n=0}^{\infty} d_n \tau_2^n \right|^2 \\ &\leq \left(\sum_{n=0}^{\infty} |d_n|^2 \right) \left(\sum_{n=0}^{\infty} |\tau_2|^{2n} \right) \\ &= \frac{\|F\|_{H^2}^2}{1 - |\tau_2|^2}. \end{aligned} \quad (\text{S253})$$

Using Eq. (S245), we obtain

$$h_\chi \leq \|F\|_{H^2}^2 = 1 - \rho^2|\tau_1|^2 \leq 1. \quad (\text{S254})$$

The canonical expression (S241) is invariant under complex conjugation combined with exchange of the two insertion points. Applying the same argument after this exchange gives

$$h_\chi \leq 1 - \rho^2|\tau_2|^2 \leq 1. \quad (\text{S255})$$

Since h_χ is nonnegative by definition,

$$0 \leq h_\chi(t_1, t_2; t_{\text{ev}}) \leq 1. \quad (\text{S256})$$

This proves Eq. (S67). Together with Eq. (S64), it establishes Eq. (S69).

References

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- [2] Lensky, Y. D. & Qi, X.-L. Rescuing a black hole in the large- q coupled SYK model. *Journal of High Energy Physics* **2021**, 116 (2021).